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2015.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

ENGINE CONTROL MODULE (ECM) - 4.4L DIESEL

**CAUTION:**

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

**NOTES:**

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Engine Control Module (ECM) - 4.4L Diesel. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Electronic Engine Controls - 4.4L V8 - TdV8 \(303-14E Electronic Engine Controls - TDV8 4.4L Diesel, Diagnosis and Testing\)](#).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1087-93	LIN Bus "A" - No operation	Generator LIN bus communication circuit failure	Refer to the electrical circuit diagrams and check the generator LIN bus circuit, for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest
B10A2-32	Crash Input - Signal low time < minimum	 NOTE: Circuit reference INI - This DTC is set when the engine control module detects that the signal low time is less than the minimum value - Restraints control module failure - Auxiliary junction box failure - Harness failure	- This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module - Check the restraints control module for DTCs and refer to the relevant DTC index - Check auxiliary junction box for DTCs and refer to the relevant DTC index - Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest
B10A2-35	Crash Input - Signal high time > maximum	 NOTE: Circuit reference INI - This DTC is set when the engine control module detects that the signal high time is greater than the maximum value - Restraints control module failure - Auxiliary junction box failure - Harness failure	- This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module - Check the restraints control module for DTCs and refer to the relevant DTC index - Check auxiliary junction box for DTCs and refer to the relevant DTC index - Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest

B10A2-36	Crash Input - Signal frequency too low	 NOTE: Circuit reference INI	▪ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module ▪ Check the restraints control module for DTCs and refer to the relevant DTC index ▪ Check auxiliary junction box for DTCs and refer to the relevant DTC index ▪ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest
B10A2-37	Crash Input - Signal frequency too high	 NOTE: Circuit reference INI	▪ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module ▪ Check the restraints control module for DTCs and refer to the relevant DTC index ▪ Check auxiliary junction box for DTCs and refer to the relevant DTC index ▪ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest
B1206-68	Crash Occurred - Event information	▪ This DTC is set if the restraints control module has deployed the restraints systems following activation of the crash sensors ▪ Event information. The engine control module has received a crash signal from the restraints control module	▪ Using the manufacturer approved diagnostic system check the restraints control module for DTCs and refer to the relevant DTC index
P0030-11	HO2S Heater Control Circuit	 NOTE:	▪ Using the manufacturer approved diagnostic system check datalogger

	(Bank 1, Sensor 1) - Circuit short to ground	Circuit reference LPPH_A ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to ground ■ Heated oxygen sensor failure	signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P0030-12	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit short to battery	 NOTE: Circuit reference LPPH_A ■ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to power ■ Heated oxygen sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P0030-13	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit open	 NOTE: Circuit reference LPPH_A ■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the	■ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required

		state of an input in response to an output ■ Harness failure - Heated oxygen sensor heater control circuit open circuit ■ Heated oxygen sensor failure	
P0030-4B	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Over temperature	■ The engine control module detected an internal temperature above the expected range ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to power ■ Heated oxygen sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P0033-00	Turbocharger /Supercharger Bypass Valve Control Circuit / Open - No sub type information	 NOTE: Circuit reference CRV ■ Harness failure - Boost air recirculation solenoid control circuit open circuit ■ Boost air recirculation solenoid failure	■ Refer to the electrical circuit diagrams and check the boost air recirculation solenoid control circuit between the engine control module and the control valve for open circuit. Check the power supply to the control valve. Repair harness as required. Clear DTC and retest ■ Check and install new boost air recirculation solenoid as required
P0034-00	Turbocharger /Supercharger Bypass Valve Control Circuit Low - No sub type information	 NOTE: Circuit reference CRV ■ Harness failure - Boost air recirculation solenoid circuit short circuit to ground ■ Boost air recirculation solenoid failure	■ Refer to the electrical circuit diagrams and check the boost air recirculation solenoid circuit between the engine control module and the control valve for a short circuit to ground. Check the power supply to the control valve. Repair harness as required. Clear DTC and retest ■ Check and install new boost air recirculation solenoid as required
P0035-00	Turbocharger /Supercharger Bypass Valve Control Circuit	 NOTE: Circuit reference CRV	■ Refer to the electrical circuit diagrams and check the boost air recirculation solenoid circuit between the engine control module and the control valve for a short circuit to power. Check the power supply to the control valve.

	High - No sub type information	■ Harness failure - Boost air recirculation solenoid circuit short circuit to power ■ Boost air recirculation solenoid failure	Repair harness as required. Clear DTC and retest ■ Check and install new boost air recirculation solenoid as required
P0039-4B	Turbocharger /Supercharger Bypass Valve Control Circuit Range /Performance - Over temperature	■ The engine control module detected an internal temperature above the expected range ■ Harness failure - Boost air recirculation solenoid circuit short circuit to ground, short circuit to power, high resistance ■ Boost air recirculation solenoid failure	■ Refer to the electrical circuit diagrams and check the boost air recirculation solenoid circuit between the engine control module and the control valve for a short circuit to ground, short circuit to power, high resistance. Check the power supply to the control valve. Repair harness as required. Clear DTC and retest ■ Check and install new boost air recirculation solenoid as required
P0045-13	Turbocharger /Supercharger Boost Control A Circuit / Open - Circuit open	■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ■ Variable geometry turbine vane actuator circuit open circuit ■ Variable geometry turbine vane actuator circuit open circuit ■ Variable geometry turbine vane actuator failure	■ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for open circuit ■ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for open circuit ■ Check and install new variable geometry turbine vane actuator as required
P0046-16	Turbocharger /Supercharger Boost Control A Circuit Range /Performance - Circuit voltage below threshold	■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Variable geometry turbine vane actuator circuit short circuit to ground ■ Variable geometry turbine vane actuator circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground ■ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground ■ Check and install new variable geometry turbine vane actuator as required

		Variable geometry turbine vane actuator failure	
P0046-19	Turbocharger /Supercharger Boost Control A Circuit Range /Performance - Circuit current above threshold	 NOTE: Variable geometry turbine vane actuator circuit consists of VGT+ VGT SENSND_19 TVP V_V5V6 - The engine control module has measured current flow above a specified range - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator failure - Engine control module failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required - Check and install new engine control module as required
P0046-1D	Turbocharger /Supercharger Boost Control A Circuit Range /Performance circuit - Current out of range	- The engine control module has detected a current outside of the expected range, but not identified as too high or too low - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required
P0046-4B	Turbocharger /Supercharger Boost Control A Circuit Range /Performance - Over temperature	 NOTE: Variable geometry turbine vane actuator circuit consists of	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required

		VGT+ VGT SENSGND_19 TVP V_V5V6 ▪ The engine control module detected an internal temperature above the expected range ▪ Variable geometry turbine vane actuator circuit short circuit to power ▪ Variable geometry turbine vane actuator failure ▪ Engine control module failure	▪ Check and install new engine control module as required
P0046-77	Turbocharger /Supercharger Boost Control A Circuit Range /Performance - Commanded position not reachable	 NOTE: Variable geometry turbine vane actuator circuit consists of VGT+ VGT SENSGND_19 TVP V_V5V6 ▪ Variable geometry turbine vane actuator circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Variable geometry turbine vane actuator failure	▪ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check for variable geometry turbine vane actuator jammed. Check and install new variable geometry turbine vane actuator as required
P0047-11	Turbocharger /Supercharger Boost Control A Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Variable geometry turbine vane actuator circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground ▪ Check and install new variable geometry turbine vane actuator as required

		- Variable geometry turbine vane actuator circuit short circuit to ground - Variable geometry turbine vane actuator failure	
P0048-12	Turbocharger /Supercharger Boost Control A Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required
P004A-13	Turbocharger /Supercharger Boost Control B Circuit / Open - Circuit open	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Turbine intake shut-off solenoid circuit open circuit - Turbine intake shut-off solenoid failure	- Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for open circuit - Check and install new turbine intake shut-off solenoid as required
P004A-71	Turbocharger /Supercharger Boost Control B Circuit / Open - Actuator stuck	- Turbine intake shut-off solenoid stuck - Turbine intake shut-off solenoid adapted end stop values outside of tolerance	- Check turbine intake shut-off solenoid for sticking - Check and install turbine intake shut-off solenoid as required
P004B-71	Turbocharger /Supercharger Boost Control B Circuit Range /Performance - Actuator stuck	Turbine intake shut-off solenoid circuit short circuit to power, short circuit to ground, open circuit	Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for short circuit to power, short circuit to ground, open circuit

		▪ Turbine intake shut-off solenoid failure	▪ Check and install new turbine intake shut-off solenoid as required
P004B-92	Turbocharger /Supercharger Boost Control B Circuit Range /Performance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Turbine intake shut-off solenoid circuit short circuit to power, short circuit to ground, open circuit ▪ Turbine intake shut-off solenoid failure	▪ Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for short circuit to power, short circuit to ground, open circuit ▪ Check and install new turbine intake shut-off solenoid as required
P004C-11	Turbocharger /Supercharger Boost Control B Circuit Low - Circuit short to ground	 NOTE: Circuit reference TSOV ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Turbine intake solenoid circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check turbine intake solenoid circuit for short circuit to ground
P004C-77	Turbocharger /Supercharger Boost Control B Circuit Low - Commanded position not reachable	 NOTE: Circuit reference TSOV ▪ Turbine intake shut-off solenoid circuit short circuit to ground ▪ Turbine intake shut-off solenoid failure	▪ Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for short circuit to ground ▪ Check and install new turbine intake shut-off solenoid as required
P004D-12	Turbocharger /Supercharger Boost Control	 NOTE:	▪ Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for short circuit to power

	B Circuit High - Circuit short to battery	Circuit reference TSOV ■ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ■ Turbine intake shut-off solenoid circuit short circuit to power ■ Turbine intake shut-off solenoid failure	■ Check and install new turbine intake shut-off solenoid as required
P004D-77	Turbocharger /Supercharger Boost Control B Circuit High - Commanded position not reachable	 NOTE: Circuit reference TSOV ■ Turbine intake shut-off solenoid circuit short circuit to power ■ Turbine intake shut-off solenoid failure	■ Refer to the electrical circuit diagrams and check the turbine intake shut-off solenoid circuit for short circuit to power ■ Check and install new turbine intake shut-off solenoid as required
P006A-00	MAP - Mass or Volume Air Flow Correlation - No sub type information	■ Intake air system, high pressure boost leak bank 1 ■ Intake air system, high pressure boost leak bi-turbo mode bank 2 ■ Intake air system, low pressure boost leak bank 1 ■ Intake air system, low pressure boost leak bank 2	■ If this DTC is logged with P1247-00, P0402-00 & P00BF-07, suspect intake air system, high pressure boost leak bank 1 ■ If this DTC is logged with P1247-00, suspect intake air system, high pressure boost leak bi-turbo mode bank 2 ■ If this DTC is logged with P00BE-07 & P0401-00, suspect intake air system, low pressure boost leak bank 1 ■ If this DTC is logged with P00BC-00, suspect intake air system, low pressure boost leak bank 2 ■ Using the manufacturer approved diagnostic system perform the (Turbo, EGR and air path dynamic test) routine
P006A-22	MAP - Mass or Volume Air Flow Correlation - Signal	■ The engine control module measured a signal voltage above a specified range but not necessarily a short	■ Check for other DTCs associated with intake and boost airflow control actuators, refer to the relevant DTC index and act on those DTCs first

	amplitude > maximum	circuit to power, gain too high ▪ Error path indicating whether over boost monitoring is active or not ▪ Failure in induction air circuit ▪ Failure in boost air circuit ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Manifold absolute pressure sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Mass air flow sensor failure ▪ Manifold absolute pressure sensor failure	▪ Refer to workshop manual and check intake air circuit, boost air circuit and associated control valves, intercooler and air filter for leaks, blockages, restrictions and control valve actuator malfunctions ▪ Refer to the electrical circuit diagrams and check both mass air flow sensor circuits for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check manifold absolute pressure sensor wiring circuits for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new mass air flow sensor or manifold absolute pressure sensor as required
P006A-84	MAP - Mass or Volume Air Flow Correlation Bank 1 - Signal below allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Intake air induction system leakage ▪ High pressure exhaust gas recirculation valve - Exhaust gas leakage into air intake system ▪ High pressure exhaust gas recirculation valve mechanical integrity ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground ▪ Manifold absolute pressure and temperature sensor	▪ Prioritised Checks to Perform ▪ Check air induction system for leakage ▪ Check high pressure exhaust gas recirculation valve for exhaust gas leakage into air intake system ▪ Check high pressure exhaust gas recirculation valve for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground ▪ Inspect manifold absolute pressure and temperature sensor connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new manifold absolute pressure and temperature sensor, only when diagnosed as failed ▪ Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground ▪ Inspect mass air flow sensor connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new mass air flow sensor, only when diagnosed as failed

		connector is disconnected, connector pin is backed out, connector pin corrosion - Manifold absolute pressure and temperature sensor failure - Mass air flow sensor circuit short circuit to ground - Mass air flow sensor connector is disconnected, connector pin is backed out, connector pin corrosion - Mass air flow sensor failure	Using the Jaguar Land Rover approved diagnostic equipment, clear all stored DTCs and retest
P0070-16	Ambient Air Temperature Sensor Circuit - Circuit voltage below threshold	 NOTE: Circuit reference AAT - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Harness failure - Ambient air temperature sensor circuit - Ambient air temperature sensor failure	- Using the manufacturer approved diagnostic system check datalogger signals, Ambient Air Temperature Sensor Voltage (0x03BA), Ambient Air Temperature (0xF466). This DTC is set when the ambient air temperature sensor signal line voltage at the engine control module is less than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Repair harness as required. Clear DTC and retest - Check and install new ambient air temperature sensor as required
P0070-17	Ambient Air Temperature Sensor Circuit - Circuit voltage above threshold	 NOTE: Circuit reference AAT - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Ambient air	- Using the manufacturer approved diagnostic system check datalogger signals, Ambient Air Temperature Sensor Voltage (0x03BA), Ambient Air Temperature (0xF466). This DTC is set when the ambient air temperature sensor signal line voltage at the engine control module is greater than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Repair harness as required. Clear DTC and retest - Check and install new ambient air temperature sensor as required

		temperature sensor circuit ▪ Ambient air temperature sensor failure	
P007A-16	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Circuit voltage below threshold	 NOTE: Circuit reference ACT ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Air charge temperature sensor circuit ▪ Air charge temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Charge Air Temperature Voltage (0x03EE). This DTC is set when the air charge temperature sensor signal line voltage at the engine control module is less than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Repair harness as required. Clear DTC and retest ▪ Check and install new charge air temperature sensor as required
P007A-17	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Circuit voltage above threshold	 NOTE: Circuit reference ACT ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Air charge temperature sensor circuit ▪ Air charge temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Air Charge Temperature (0x051C). This DTC is set when the air charge temperature sensor signal line voltage at the engine control module is greater than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Repair harness as required. Clear DTC and retest ▪ Check and install new charge air temperature sensor as required
P007A-62	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Signal compare failure	▪ Harness failure - Air charge temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Air charge temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Charge Air Temperature Voltage (0x03EE). Refer to the electrical circuit diagrams and check air charge temperature sensor circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new charge air temperature sensor as required

P0087-00	Fuel Rail /System Pressure - Too Low - No sub type information	▪ Low level fuel condition ▪ Fuel gauge sender unit sticking ▪ Fuel lines restricted ▪ Fuel lines leaking ▪ Fuel pump failure ▪ Fuel rail pressure sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel rail pressure sensor failure	▪ Check the fuel volume available in the fuel tank is sufficient, add fuel as required. Clear DTC and retest ▪ Check the fuel gauge sender units for correct operation ▪ Check fuel lines for restriction ▪ Check fuel lines for leakage ▪ Check for fuel pump related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair harness as required ▪ Check and install new fuel rail pressure sensor as required
P0087-16	Fuel Rail /System Pressure - Too Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Fuel rail pressure sensor circuit, short circuit to ground, open circuit, high resistance ▪ Fuel lines leaking ▪ Fuel lines restricted ▪ Fuel rail pressure sensor failure	▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair harness as required. Clear DTC and retest ▪ Check the fuel lines for any signs of leakage, replace as required ▪ Check the fuel lines for any signs of damage, crimping or restrictions, replace as required ▪ Check and install new fuel rail pressure sensor as required
P0087-21	Fuel Rail /System Pressure - Too Low - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Fuel rail pressure sensor signal circuit short circuit to ground, open circuit ▪ Fuel rail pressure sensor voltage reference circuit high resistance, open circuit ▪ Fuel lines leaking ▪ Fuel lines restricted ▪ Fuel rail pressure sensor internal failure	▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor signal circuit for short circuit to ground, open circuit ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor voltage reference circuit for high resistance, open circuit ▪ Check the fuel lines for any signs of leakage, replace as required ▪ Check the fuel lines for any signs of damage, crimping or restrictions, replace as required ▪ Using the manufacturer approved diagnostic system check for fuel pump related DTCs and refer to the relevant DTC index. Check and install new fuel rail pressure sensor as required ▪ Check and install new fuel pump as required

		Fuel pump internal failure	
P0088-00	Fuel Rail /System Pressure - Too High - No sub type information	- Fuel gauge sender unit sticking - Fuel rail pressure sensor signal circuit short circuit to power - Fuel rail pressure sensor ground circuit open circuit, high resistance - Fuel pressure control valve circuit short circuit to power, ground, open circuit, high resistance - Fuel volume control valve circuit short circuit to power, ground, open circuit, high resistance - Fuel rail pressure sensor failure - Fuel pressure control valve failure - Fuel volume control valve failure	- Check the fuel gauge sender units for correct operation - Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power. Repair harness as required - Refer to the electrical circuit diagrams and check fuel rail pressure sensor ground circuit for open circuit, high resistance. Repair harness as required - Refer to the electrical circuit diagrams and check fuel pressure control valve circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required - Refer to the electrical circuit diagrams and check fuel volume control valve circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required - Check and install new fuel rail pressure sensor as required - Check and install new fuel pressure control valve as required - Check and install new fuel volume control valve as required
P0088-17	Fuel Rail /System Pressure - Too High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Fuel rail pressure sensor signal circuit short circuit to power	Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power
P0088-22	Fuel Rail /System Pressure - Too High - Signal amplitude > maximum	- The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high - Low level fuel condition - Fuel gauge sender unit sticking - Fuel rail pressure sensor signal circuit short circuit to power - Fuel rail pressure sensor internal failure	- Check the fuel volume available in the fuel tank is sufficient, add fuel as required - Check the fuel gauge sender units for correct operation - Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to power - Check and install new fuel rail pressure sensor as required

P0089-21	Fuel Pressure Regulator Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Fuel pressure regulator signal circuit short circuit to ground, open circuit ▪ Fuel pressure regulator reference voltage circuit high resistance ▪ Fuel pressure regulator failure	▪ Refer to the electrical circuit diagrams and check fuel pressure regulator circuit for short circuit to ground, open circuit. Repair harness as required ▪ Refer to the electrical circuit diagrams and check fuel pressure regulator reference voltage circuit for high resistance. Repair harness as required ▪ Check and install new fuel pressure regulator as required
P0089-22	Fuel Pressure Regulator Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Fuel pressure regulator signal circuit short circuit to power ▪ Fuel pressure regulator failure	▪ Refer to the electrical circuit diagrams and check fuel pressure regulator circuit for short circuit to power. Repair harness as required ▪ Check and install new fuel pressure regulator as required
P0090-13	Fuel Pressure Regulator 1 Control Circuit /Open - Circuit open	 NOTE: Circuit reference PCV ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Fuel pressure control valve control circuit open circuit	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). Refer to the electrical circuit diagrams and check the fuel pressure control valve control circuit for open circuit. Repair harness as required. Clear DTC and retest
P0090-4B	Fuel Pressure Regulator 1 Control Circuit /Open - Over temperature	▪ The engine control module detected an internal temperature above the expected range	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). Refer to the electrical circuit diagrams and check fuel pressure control valve

		■ Fuel pressure control valve signal circuit short circuit to power, open circuit ■ Fuel pressure regulator failure	circuit for short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new fuel pressure regulator as required
P0091-11	Fuel Pressure Regulator 1 Control Circuit Low - Circuit short to ground	 NOTE: Circuit reference PCV ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Fuel pressure control valve circuit short circuit to ground ■ Fuel pressure control valve failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). Refer to the electrical circuit diagrams and check the fuel pressure control valve circuit for short circuit to ground. Check power supply circuit to control valve. Repair harness as required. Clear DTC and retest ■ Check and install new fuel pressure control valve as required
P0091-16	Fuel Pressure Regulator 1 Control Circuit Low - Circuit voltage below threshold	 NOTE: Circuit reference PCV ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Harness failure - Fuel pressure control valve circuit ■ Fuel pressure control valve failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). This DTC is set when the voltage on the fuel pressure control valve circuit is less than the threshold expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel pressure control valve circuit for high resistance, short circuit to ground. Check power supply circuit to the fuel pressure control valve. Repair harness as required. Clear DTC and retest ■ Check and install new fuel pressure control valve as required
P0092-12	Fuel Pressure Regulator 1 Control Circuit High - Circuit short to battery	 NOTE: Circuit reference PCV	■ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). Refer to the electrical circuit diagrams and check the fuel pressure control valve circuit for short circuit to power. Check power supply circuit to control

		▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Fuel pressure control valve circuit short circuit to power	valve. Repair harness as required. Clear DTC and retest
P0092-17	Fuel Pressure Regulator 1 Control Circuit High - Circuit voltage above threshold	 NOTE: Circuit reference PCV ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Fuel pressure control valve circuit ▪ Fuel pressure control valve failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Pressure Relief Control Valve Duty Cycle (0x03C3). This DTC is set when the voltage on the fuel pressure control valve circuit is more than the threshold expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel pressure control valve circuit for high resistance, short circuit to power. Check power supply circuit to the fuel pressure control valve. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel pressure control valve as required
P00BC-00	Mass or Volume Air Flow A Circuit Range /Performance - Air Flow Too Low - No sub type information	▪ Boost air solenoid stuck shut bi-turbo mode ▪ Turbine intake solenoid stuck shut ▪ Intake system low pressure boost leak bank 2	▪ If this DTC is logged with P1247-00, suspect boost air solenoid stuck shut bi-turbo mode ▪ If this DTC is logged with P22D3-77 & P22CF-71, suspect turbine intake solenoid stuck shut ▪ If this DTC is logged with P006A-00, suspect intake system low pressure boost leak bank 2
P00BC-07	Mass or Volume Air Flow A Circuit Range /Performance - Air Flow Too Low - Mechanical failures	▪ Air flow disruption at sensing element of mass air flow sensor ▪ Mass air flow sensor failure	▪ This DTC is set if the mass air flow signal from sensor 1 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt air flow at the sensing element. Clear DTC and retest ▪ Check and install new mass air flow sensor as required

P00BE-00	Mass or Volume Air Flow B Circuit Range /Performance - Air Flow Too Low - No sub type information	- Boost air recirculation solenoid stuck open bi-turbo mode - Air flow disruption at sensing element of mass air flow sensor 2 - Harness failure - Mass air flow sensor circuit short circuit to ground, high resistance, open circuit - Mass air flow sensor 2 failure	- If this DTC is logged with P1247-00, suspect boost air recirculation solenoid stuck open. Refer to the relevant sections of the workshop manual and check the boost air recirculation solenoid circuit for short circuit to power, short circuit to ground, open circuit. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element - Check and install new boost air recirculation solenoid as required - Refer to the electrical circuit diagrams and check mass air flow sensor 2 for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new mass air flow sensor 2 as required
P00BE-07	Mass or Volume Air Flow B Circuit Range /Performance - Air Flow Too Low - Mechanical failures	- Air flow disruption at sensing element of mass air flow - Mass air flow failure	- This DTC is set if the mass air flow signal from sensor 2 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element. Clear DTC and retest - Check and install new mass air flow as required
P00BF-07	Mass or Volume Air Flow B Circuit Range /Performance - Air Flow Too High - Mechanical failures	Intake air system, high pressure boost leak	If this DTC is logged with P1247-00, P006A-00 & P00BF-07, suspect intake air system, high pressure boost leak
P00CF-62	Secondary Compressor Outlet Pressure Sensor - Signal compare failure	- Boost air circuit leak or blockage - Boost air recirculation solenoid failure - Boost air solenoid failure - Mechanical failure - Sensor hose blocked or leaking	- This DTC is set when the engine control module detects a signal compare failure on the boost air pressure sensor signal. Check the sensor hose for blockages or leaks. Check the sensor and harness for mechanical damage caused by heat or chaffing - Refer to the electrical circuit diagrams and check the secondary compressor

		■ Harness failure - Boost air pressure sensor circuit ■ Boost air pressure sensor failure	outlet pressure sensor power and ground supplies for short circuit to ground, open circuit. Check the signal line for short circuit to ground, short circuit to power, open circuit. Repair harness as required ■ Check and install new secondary compressor outlet pressure sensor as required
P0102-00	Mass or Volume Air Flow A Circuit Low - No sub type information	 NOTE: Circuit reference MAF_A ■ Air flow disruption at sensing element of mass air flow sensor ■ Harness failure - Mass air flow sensor circuits ■ Mass air flow sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Air Flow Rate From Mass Air Flow Sensor Bank 1 (0x0504). This DTC is set if the mass air flow signal from sensor 1 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element ■ Refer to the electrical circuit diagrams and check the mass air flow sensor power and ground circuits for high resistance, short circuit to power, short circuit to ground. Check the mass air flow signal line for high resistance. Repair harness as required. Clear DTC and retest ■ Check and install new mass air flow sensor as required
P0103-00	Mass or Volume Air Flow A Circuit High - No sub type information	 NOTE: Circuit reference MAF_A ■ Air flow disruption at sensing element of mass air flow sensor ■ Harness failure - Mass air flow sensor circuits ■ Mass air flow sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Air Flow Rate From Mass Air Flow Sensor Bank 1 (0x0504). This DTC is set if the mass air flow signal from sensor 1 is greater than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element. ■ Refer to the electrical circuit diagrams and check the mass air flow sensor power and ground circuits for high resistance, short circuit to power, short circuit to ground. Check the mass air flow signal line for high resistance. Repair harness as required. Clear DTC and retest ■ Check and install new mass air flow sensor as required

P0105-16	Manifold Absolute Pressure/BARO circuit - Circuit voltage below threshold	 NOTE: Circuit reference MAP_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold absolute pressure sensor	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for open circuit, high resistance, short circuit to ground. Check the sensor ground for open circuit, short circuit to power. Check the signal circuit for short circuit to ground, high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new manifold absolute pressure sensor as required ▪ Using the manufacturer approved diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0322), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the voltage on the manifold absolute pressure sensor signal circuit is less than the threshold expected by the engine control module. Refer to the workshop manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor
P0105-17	Manifold Absolute Pressure/BARO circuit - Circuit voltage above threshold	 NOTE: Circuit reference MAP_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold pressure sensor	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for short circuit to power. Check the sensor ground for open circuit, short circuit to power. Check the signal circuit for short circuit to power. Repair harness as required. Clear DTC and retest ▪ Check and install new manifold absolute pressure sensor as required ▪ Using the manufacturer approved diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the voltage on the manifold absolute pressure sensor signal circuit is greater than the threshold expected by the engine control module. Refer to the workshop manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor
P0107-00	Manifold Absolute	 NOTE:	▪ Refer to the workshop manual and check the manifold absolute pressure

	Pressure/BARO Sensor Low - No sub type information	Circuit reference MAP_A - Blockage or leak between the intake manifold and the manifold absolute pressure sensor short circuit to ground, short circuit to power, open circuit, high resistance - Manifold absolute pressure sensor circuit - Manifold absolute pressure sensor failure	sensor is correctly connected to the air intake manifold and there are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor - Refer to the electrical circuit diagrams and check the manifold absolute pressure sensor 5 volt supply circuit for short circuit to ground short circuit to power, open circuit, high resistance. Check the sensor ground for short circuit to ground, short circuit to power, open circuit, high resistance. Repair wiring as required, clear the DTC and retest the system - Check and install new manifold absolute pressure sensor as required
P0108-00	Manifold Absolute Pressure/BARO Sensor High - No sub type information	 NOTE: Circuit reference MAP_A - Blockage or leak between the intake manifold and the manifold absolute pressure sensor short circuit to ground, short circuit to power, open circuit, high resistance - Manifold absolute pressure sensor circuit - Manifold absolute pressure sensor failure	- Refer to the workshop manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor - Refer to the electrical circuit diagrams and check the manifold absolute pressure sensor 5 volt supply circuit for short circuit to ground short circuit to power, open circuit, high resistance. Check the sensor ground for short circuit to ground, short circuit to power, open circuit, high resistance. Repair wiring as required, clear the DTC and retest the system - Check and install new manifold absolute pressure sensor as required
P010C-00	Mass or Volume Air Flow B Circuit Low - No sub type information	 NOTE: Circuit reference MAF_B - Air flow disruption at sensing element of mass air flow - Harness failure - Mass air flow sensor circuits - Mass air flow failure	- Using the manufacturer approved diagnostic system check datalogger signals, Air Flow Rate From Mass Air Flow Sensor Bank 2 (0x0505). This DTC is set if the mass air flow signal from sensor 2 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element - Refer to the electrical circuit diagrams and check the power and ground circuits for the mass air flow sensors for high resistance, short circuit to power, short circuit to ground. Check the mass air flow signal line between

			the engine control module and the mass air flow sensor for high resistance. Repair harness as required. Clear DTC and retest Check and install new mass air flow sensor as required
P010D-00	Mass or Volume Air Flow B Circuit High - No sub type information	 NOTES: - Circuit reference MAF_B - Air flow signal too high on secondary turbine bank - Air flow disruption at sensing element of mass air flow sensor - Harness failure - Mass air flow sensor circuits - Mass air flow sensor failure	- Using the manufacturer approved diagnostic system check datalogger signals, Air Flow Rate From Mass Air Flow Sensor Bank 2 (0x0505). This DTC is set if the mass air flow signal from sensor B is greater than the value expected by the Engine Control Module. Refer to the relevant sections of the workshop manual and check the induction system for air leaks, and obstructions to flow. Check the condition of the air filter and examine the induction pipes for debris which could disrupt airflow at the sensing element - Refer to the electrical circuit diagrams and check the power and ground circuits for the mass air flow sensors for high resistance, short circuits to power or ground, check the mass air flow signal line between the engine control module and the mass air flow sensor for high resistance, and intermittent failures. Repair wiring as required, clear the DTC and retest the system - Check and install new mass air flow sensor as required
P0110-16	Intake Air Temperature Sensor 1 Circuit (Bank 1) - Circuit voltage below threshold	 NOTE: Circuit reference IAT - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Intake air temperature sensor located in mass air flow sensor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the intake air temperature sensor located in mass air flow sensor circuit for short circuit to ground
P0110-17	Intake Air Temperature Sensor 1 Circuit (Bank 1) - Circuit	 NOTE: Circuit reference IAT	Refer to the electrical circuit diagrams and check the intake air temperature sensor located in mass air flow sensor circuit for short circuit to power

	voltage above threshold	 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Intake air temperature sensor located in mass air flow sensor circuit short circuit to power	
P0116-00	Engine Coolant Temperature Sensor 1 Circuit Range /Performance - No sub type information	 NOTE: Circuit reference ECT_A ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DTC is set if the engine coolant temperature value fails plausibility checks by the engine control module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, short circuit to other circuits. Repair harness as required. Clear DTC and retest ▪ Check and install new engine coolant temperature sensor as required
P0116-16	Engine Coolant Temperature Sensor 1 Circuit Range /Performance - Circuit voltage below threshold	 NOTE: Circuit reference ECT_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DTC is set if the engine coolant temperature sensor signal voltage is less than the value expected by the engine control module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, short circuit to other circuits. Repair harness as required. Clear DTC and retest ▪ Check and install new engine coolant temperature sensor as required
P0116-	Engine Coolant		

17	Temperature Sensor 1 Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference ECT_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DTC is set if the engine coolant temperature sensor signal voltage is greater than the value expected by the control module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, short circuit to other circuits. Repair harness as required. Clear DTC and retest ▪ Check and install new engine coolant temperature sensor as required
P0116-21	Engine Coolant Temperature Sensor 1 Circuit Range /Performance - Signal amplitude < minimum	 NOTE: Circuit reference ECT_A ▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuit short circuit to ground, high resistance, open circuit ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DTC is set if the engine coolant temperature sensor signal amplitude is lower than the value expected by the control module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine coolant temperature sensor as required
P0116-22	Engine Coolant Temperature Sensor 1 Circuit Range /Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Engine coolant level or flow failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DTC is set if the engine coolant temperature sensor signal amplitude is greater than the value expected by the control module. Refer to the workshop

		■ Harness failure - Engine coolant temperature sensor circuit short circuit to power ■ Engine coolant temperature sensor failure	manual and check the engine cooling system to ensure the coolant condition and level is correct ■ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new engine coolant temperature sensor as required
P0130-11	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit short to ground	 NOTE: Circuit reference LPV_A ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Heated oxygen sensor circuit short circuit to ground ■ Heated oxygen sensor component failure	■ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P0130-12	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit short to battery	 NOTE: Circuit reference LPV_A ■ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ■ Harness failure - Heated oxygen sensor circuit short circuit to power ■ Heated oxygen sensor component failure	■ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required

P0130-13	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit open	 NOTE: Circuit reference LPPC_A LPV_A	- Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P0130-26	O2 Sensor Circuit (Bank 1 Sensor 1) - Signal rate of change below threshold	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure - Heated oxygen sensor circuit open circuit - Heated oxygen sensor component failure	- Check for and rectify any exhaust leak between cylinder head and catalytic converter. Check heated oxygen sensor is correctly installed in exhaust manifold - Check fuel control system for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check heated oxygen sensor to engine control module circuit for short circuit to ground, short circuit to power, high resistance. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P0133-00	O2 Sensor Circuit Slow Response (Bank 1 Sensor 1) - No sub type information	- Exhaust system leak - Fuel control system failure - Heated oxygen sensor to engine control module wiring shield high resistance - Heated oxygen sensor failure	- Check for and rectify any exhaust leak between cylinder head and catalytic converter. Check heated oxygen sensor is correctly installed in exhaust manifold - Check fuel control system for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check heated oxygen sensor to engine control module wiring shield for high resistance. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required

P0148-00	Fuel Delivery Error - No sub type information	High pressure fuel pump damage caused by excessive accumulated pressure	- Clear DTC and re-test. Check connections are secure, check wiring integrity - Check and install new High pressure fuel pump as required
P0181-16	Fuel Temperature Sensor A Circuit Range /Performance - Circuit voltage below threshold	 NOTE: Circuit reference IFTS - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Harness failure - Fuel temperature sensor circuit failure - Sensor component failure	- Using the manufacturer approved diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fuel Rail Temperature Sensor Voltage (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is less than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal circuit for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new fuel temperature sensor as required
P0181-17	Fuel Temperature Sensor A Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference IFTS - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Fuel temperature sensor circuit failure - Sensor component failure	- Using the manufacturer approved diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fuel Rail Temperature Sensor Voltage (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is greater than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal and ground circuits for short circuit to power, open circuit or high resistance. Repair harness as required. Clear DTC and retest - Check and install new fuel temperature sensor as required
P0182-00	Fuel Temperature Sensor A Circuit Low - No sub type information	 NOTE: Circuit reference IFTS Harness failure - Fuel temperature sensor circuit failure	Using the manufacturer approved diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fuel Rail Temperature Sensor Voltage (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is less than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal circuit for short circuit to ground. Repair harness as required. Clear DTC and retest

		▪ Sensor component failure	▪ Check and install new fuel temperature sensor as required
P0183-00	Fuel Temperature Sensor A Circuit High - No sub type information	 NOTE: Circuit reference IFTS ▪ Harness failure - Fuel temperature sensor circuit failure ▪ Sensor component failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fuel Rail Temperature Sensor Voltage (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is greater than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal and ground circuits for short circuit to power, open circuit or high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel temperature sensor as required
P0192-16	Fuel Rail Pressure Sensor A Circuit Low - Circuit voltage below threshold	 NOTE: Circuit reference RPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor circuit failure ▪ Sensor component failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressure transducer mounted on the fuel rail. This DTC is set when the voltage on the signal line is less than that expected by the engine control module. Refer to the workshop manual and ensure that the fuel supply system is working correctly ▪ Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for short circuit to power, short circuit to ground, open circuit, high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel rail pressure sensor as required
P0193-17	Fuel Rail Pressure Sensor A Circuit High - Circuit voltage above threshold	 NOTE: Circuit reference RPS ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressure transducer mounted on the fuel rail. This DTC is set when the voltage on the signal line is greater than that expected by the engine control module. Refer to the workshop manual and ensure that the fuel supply system is working correctly ▪ Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for short circuit to power, short circuit to

		▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor circuit failure ▪ Sensor component failure	ground, open circuit, high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel rail pressure sensor as required
P0194-00	Fuel Rail Pressure Sensor A Circuit Intermittent /Erratic - No sub type information	▪ Fuel supply system failure ▪ Fuel rail pressure sensor A circuit high resistance ▪ Fuel temperature sensor A failure ▪ Fuel lift pump failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423) ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor A circuit for high resistance ▪ Using the manufacturer approved diagnostic system check for related fuel system DTCs ▪ Check and install new fuel rail pressure A sensor as required ▪ Check and install new fuel lift pump as required
P0195-00	Engine Oil Temperature Sensor Circuit - No sub type information	 NOTE: Circuit reference OTL ▪ Oil contaminated or level incorrect ▪ Harness failure - Oil level and temperature sensor circuits ▪ Oil level and temperature sensor failure	▪ Check the oil level is correct and the oil does not appear contaminated. Renew or top up oil as required ▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line for open circuit, short circuit to power, short circuit to ground, high resistance and intermittent connections. Repair harness as required. Clear DTC and retest ▪ Check and install new engine oil level and temperature sensor as required
P0195-23	Engine Oil Temperature Sensor Circuit - Signal stuck low	 NOTE: Circuit reference OTL ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Oil contaminated or level incorrect	▪ Check the oil level is correct and the oil does not appear contaminated. Renew or top up oil as required ▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line for open circuit, short circuit to power, short circuit to ground, high resistance and intermittent connections. Repair harness as required. Clear DTC and retest ▪ Check and install new engine oil level and temperature sensor as required

		■ Harness failure - Oil level and temperature sensor circuits ■ Oil level and temperature sensor failure	
P0195-62	Engine Oil Temperature Sensor Circuit - Signal compare failure	 NOTE: Circuit reference OTL ■ Oil contaminated or level incorrect ■ Harness failure - Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ■ Oil level and temperature sensor failure	■ Check the oil does not appear contaminated and the level is correct. Renew or top up oil as required ■ Refer to the electrical circuit diagrams and check oil level and temperature sensor signal circuit for short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ■ Start the engine from cold and allow to idle, check and record Sump Oil Temperature - Measured (0x03F3) datalogger signal. Continue to warm up at idle, after approximately 10 minutes check and record Sump Oil Temperature - Measured (0x03F3) signal. If value of signal has not increased by 5°C check and install new oil level and temperature sensor as required
P0201-00	Cylinder 1 Injector Circuit /Open - No sub type information	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ■ Harness failure - Cylinder 1 injector circuit open circuit ■ Cylinder 1 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 1 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0202-00	Cylinder 2 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ■ Harness failure - Cylinder 2 injector circuit open circuit ■ Cylinder 2 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 2 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required

P0203-00	Cylinder 3 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23 ■ Harness failure - Cylinder 3 injector circuit open circuit ■ Cylinder 3 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 3 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0204-00	Cylinder 4 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12 ■ Harness failure - Cylinder 4 injector circuit open circuit ■ Cylinder 4 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 4 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0205-00	Cylinder 5 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21 ■ Harness failure - Cylinder 5 injector circuit open circuit ■ Cylinder 5 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 5 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0206-00	Cylinder 6 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ■ Harness failure - Cylinder 6 injector circuit open circuit	■ This DTC is set when the engine control module detects an open circuit on the cylinder 6 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required

		■ Cylinder 6 injector failure	
P0207-00	Cylinder 7 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 ■ Harness failure - Cylinder 7 injector circuit open circuit ■ Cylinder 7 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 7 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0208-00	Cylinder 8 Injector Circuit / Open - No sub type information	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 ■ Harness failure - Cylinder 8 injector circuit open circuit ■ Cylinder 8 injector failure	■ This DTC is set when the engine control module detects an open circuit on the cylinder 8 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector for high resistance, open circuits or intermittent connections. Repair harness as required ■ Check and install new fuel injector as required
P0216-00	Injector /Injection timing Control Circuit - No sub type information	■ Internal engine control module power supply is unable to supply the fuel injectors with the maximum number of injections required ■ Internal engine control module monitoring has detected a fuel pressure not able to meet the demand for the number of fuel injections ■ At high engine speeds the internal engine control module monitoring has detected that the computing time available for the desired number of injections is not sufficient	■ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ■ Check the vehicle charging system performance to ensure the voltage regulation is correct ■ Refer to the electrical circuit diagrams and check engine control module power and ground circuits ■ Check fuel control system for related DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, clear DTC and retest
P0219-	Engine		

00	Overspeed Condition - No sub type information	- Camshaft or crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit - Camshaft or crankshaft position sensor failure	- Refer to the electrical circuit diagrams and check camshaft and crankshaft position sensor circuits for short circuit to ground, short circuit to power, open circuit - Check for engine oil ingestion to sensors. Check and install new camshaft and crankshaft position sensor as required
P0235-16	Turbocharger /Supercharger Boost Sensor A Circuit - Circuit voltage below threshold	 NOTE: Circuit reference SCOP - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Mechanical failure - Sensor hose blocked or leaking - Fault affecting intake air circuit or boost air circuit control valves or actuators - Harness failure - Boost air pressure sensor circuit - Boost air pressure sensor failure	- Using the manufacturer approved diagnostic system check datalogger signals, Turbocharger/Supercharger Boost Sensor A Circuit (0x033C), Boost Absolute Pressure - Raw Value (0x033E). This DTC is set when the engine control module detects a signal voltage from the boost air pressure sensor signal line which is less than the threshold value. Check the sensor hose for blockages or leaks - Check the intake air and boost air circuits for failures including leaks, blockages, control valve actuator malfunctions - Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line for open circuit, short circuit to ground, high resistance and intermittent connections. Repair harness as required. Clear DTC and retest - Check and install new boost air pressure sensor as required
P0235-17	Turbocharger /Supercharger Boost Sensor A Circuit - Circuit voltage above threshold	 NOTE: Circuit reference SCOP - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Mechanical failure - Sensor hose blocked or leaking - Fault affecting intake air circuit or boost air circuit control valves or actuators	- Using the manufacturer approved diagnostic system check datalogger signals, Turbocharger/Supercharger Boost Sensor A Circuit (0x033C), Boost Absolute Pressure - Raw Value (0x033E). This DTC is set when the engine control module detects a signal voltage from the boost air pressure sensor signal line which is greater than the threshold value. Check the sensor hose for blockages or leaks - Check the intake air and boost air circuits for failures including leaks, blockages, control valve actuator malfunctions - Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line for open circuit, short circuit to power. Repair harness as required. Clear DTC and retest

		▪ Harness failure - Boost air pressure sensor circuit ▪ Boost air pressure sensor failure	▪ Check and install new boost air pressure sensor as required
P0235-94	Turbocharger /Supercharger Boost Sensor A Circuit - Unexpected operation	 NOTE: Circuit reference SCOP ▪ Boost air solenoid stuck open mono turbo mode ▪ Turbine intake solenoid leakage when closed ▪ Turbine intake solenoid stuck open ▪ Intake air system, blocked low pressure air intake. This failure mode can be caused by snow packing in the intake system. Symptoms often disappear after the vehicle has been warmed and heat soaked. Similar symptoms to seized primary turbo	▪ If this DTC is logged with P00BD-07 suspect, boost air solenoid stuck open mono turbo mode ▪ If this DTC is logged with P00BD-07 suspect, turbine intake solenoid leakage when closed ▪ If this DTC is logged with P00BD-07, P22D2-77, P1247-00 & P22CF-71 suspect, turbine intake solenoid stuck open ▪ If this DTC is logged with P00BD-07, P22D2-77, P1247-00 & P22CF-71 suspect, intake air system, blocked low pressure air intake ▪ Using the manufacturer approved diagnostic system perform the (Turbo, EGR and air path dynamic test) routine
P023D-00	Manifold Absolute Pressure-Turbocharger /Supercharger Boost Sensor A Correlation - No sub type information	▪ Induction system air leak or blockage ▪ Boost air system leak or blockage ▪ Manifold absolute pressure sensor A failure ▪ Variable geometry turbine vane actuator sticking, failure ▪ Turbocharger A failure	▪ Check induction system for leaks, blockages ▪ Check boost air system for leaks, blockages. Check for related DTCs and refer to the relevant DTC index ▪ Check and install new manifold absolute pressure sensor as required ▪ Check and install new variable geometry turbine vane actuator as required ▪ Check turbocharger rod connection and oil seals
P0251-13	Injection Pump Fuel Metering Control A - Circuit open	 NOTE: Circuit reference MEU	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel

		▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Fuel volume control valve circuit open circuit ▪ Fuel volume control valve failure	volume control valve for open circuit. Check power supply to fuel volume control valve. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0251-4B	Injection Pump Fuel Metering Control A - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Fuel volume control valve circuit short circuit to ground, short circuit to power, high resistance ▪ Fuel volume control valve failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel volume control valve for short circuit to ground, short circuit to power, high resistance. Check power supply to fuel volume control valve. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0252-16	Injection Pump Fuel Metering Control A Range /Performance - Circuit voltage below threshold	 NOTE: Circuit reference MEU ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Fuel volume control valve circuit ▪ Fuel volume control valve failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is less than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel volume control valve for an intermittent open circuit, high resistance, short circuit to ground. Check the power supply to fuel volume control valve. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0252-17	Injection Pump Fuel Metering Control A Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference MEU	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is greater than

		▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Fuel volume control valve circuit ▪ Fuel volume control valve failure	that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the volume control valve for a short circuit to power. Check the power supply to fuel volume control valve. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0253-00	Injection Pump Fuel Metering Control A Low - No sub type information	 NOTE: Circuit reference MEU ▪ Harness failure - Fuel volume control valve circuit short circuit to ground, open circuit ▪ Fuel volume control valve failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is less than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel volume control valve for a short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0254-00	Injection Pump Fuel Metering Control A High - No sub type information	 NOTE: Circuit reference MEU ▪ Harness failure - Fuel volume control valve circuit short circuit to power ▪ Fuel volume control valve failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is greater than that expected by the engine control module. Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel volume control valve for a short circuit to power. Repair harness as required. Clear DTC and retest ▪ Check and install new fuel volume control valve as required
P0255-00	Injection Pump Fuel Metering Control A Intermittent - No sub type information	 NOTE: Circuit reference MEU	▪ Using the manufacturer approved diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrical circuit diagrams and check the fuel volume control valve circuit between the engine control module and the fuel

		■ Harness failure - Fuel volume control valve circuit intermittent short circuit to ground, short circuit to power, high resistance ■ Fuel volume control valve intermittent failure	volume control valve for an intermittent short circuit to ground, short circuit to power, high resistance. Repair harness as required. Clear DTC and retest ■ Check and install new fuel volume control valve as required
P0261-00	Cylinder 1 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ■ Harness failure - Injector control circuit short circuit ■ Injector failure	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 1 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0261-11	Cylinder 1 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Injector control circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 1 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0261-23	Cylinder 1 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ■ The engine control module measures a signal that remains low when transitions are expected	■ Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 1 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest

		■ Harness failure - Short circuit between injector control circuits on different cylinders	
P0264-00	Cylinder 2 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ■ Harness failure - Injector control circuit short circuit ■ Injector failure	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 2 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0264-11	Cylinder 2 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Injector control circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 2 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0264-23	Cylinder 2 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ■ The engine control module measures a signal that remains low when transitions are expected ■ Harness failure - Short circuit between injector control circuits on different cylinders	■ Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 2 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest

P0267-00	Cylinder 3 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 3 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0267-11	Cylinder 3 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 3 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0267-23	Cylinder 3 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23	Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 3 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P0270-00	Cylinder 4 Injector Circuit Low - No sub	 NOTE:	Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module

	type information	Circuit reference O_P_PVL12 O_P_PVH12 ■ Harness failure - Injector control circuit short circuit ■ Injector failure	and the cylinder 4 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0270-11	Cylinder 4 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12 ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Injector control circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 4 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0270-23	Cylinder 4 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12 ■ The engine control module measures a signal that remains low when transitions are expected ■ Harness failure - Short circuit between injector control circuits on different cylinders	■ Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 4 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P0273-00	Cylinder 5 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 5 injector for short circuit to ground or short between the two wires. This circuit is a twisted

		■ Harness failure - Injector control circuit short circuit ■ Injector failure	- pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0273-11	Cylinder 5 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21 ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Injector control circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 5 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0273-23	Cylinder 5 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21 ■ The engine control module measures a signal that remains low when transitions are expected ■ Harness failure - Short circuit between injector control circuits on different cylinders	■ Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 5 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P0276-00	Cylinder 6 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ■ Harness failure - Injector control circuit short circuit	■ Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 6 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required

		Injector failure	
P0276-11	Cylinder 6 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Harness failure - Injector control circuit short circuit to ground	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 6 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0276-23	Cylinder 6 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 - The engine control module measures a signal that remains low when transitions are expected - Harness failure - Short circuit between injector control circuits on different cylinders	Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 6 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P0279-00	Cylinder 7 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 - Harness failure - Injector control circuit short circuit - Injector failure	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 7 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0279-	Cylinder 7	 NOTE:	

11	Injector Circuit Low - Circuit short to ground	Circuit reference O_P_PVL14 O_P_PVH14 - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Harness failure - Injector control circuit short circuit to ground	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 7 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0279-23	Cylinder 7 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 - The engine control module measures a signal that remains low when transitions are expected - Harness failure - Short circuit between injector control circuits on different cylinders	Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 7 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P0282-00	Cylinder 8 Injector Circuit Low - No sub type information	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 - Harness failure - Injector control circuit short circuit - Injector failure	- Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 8 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new fuel injector as required
P0282-11	Cylinder 8 Injector Circuit Low - Circuit short to ground	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24	Refer to the electrical circuit diagrams and check the injector control circuit between the engine control module and the cylinder 8 injector for short circuit to ground or short between the two wires. This circuit is a twisted pair, check both high and low sides for

		■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Harness failure - Injector control circuit short circuit to ground	short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new fuel injector as required
P0282-23	Cylinder 8 Injector Circuit Low - Signal stuck low	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 ■ The engine control module measures a signal that remains low when transitions are expected ■ Harness failure - Short circuit between injector control circuits on different cylinders	■ Refer to the electrical circuit diagrams and check the injector control circuits between the engine control module and the cylinder 8 injector for short circuit to other injector control circuits. Repair harness as required. Clear DTC and retest
P02CD-00	Cylinder 1 Fuel Injector Offset Learning at Max Limit - No sub type information	■ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ■ Fuel injector internal components out of tolerance	■ Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest ■ Check and install new a fuel injector as required
P02CF-00	Cylinder 2 Fuel Injector Offset Learning at Max Limit - No sub type information	■ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ■ Fuel injector internal components out of tolerance	■ Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest ■ Check and install new a fuel injector as required
P02D1-00	Cylinder 3 Fuel Injector Offset Learning at Max Limit - No sub type information	■ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit	■ Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest ■ Check and install new a fuel injector as required

		Fuel injector internal components out of tolerance	
P02D3-00	Cylinder 4 Fuel Injector Offset Learning at Max Limit - No sub type information	- Corrected set point voltage of the piezo actuator violates the on board diagnostic limit - Fuel injector internal components out of tolerance	- Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest - Check and install new a fuel injector as required
P02D5-00	Cylinder 5 Fuel Injector Offset Learning at Max Limit - No sub type information	- Corrected set point voltage of the piezo actuator violates the on board diagnostic limit - Fuel injector internal components out of tolerance	- Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest - Check and install new a fuel injector as required
P02D7-00	Cylinder 6 Fuel Injector Offset Learning at Max Limit - No sub type information	- Corrected set point voltage of the piezo actuator violates the on board diagnostic limit - Fuel injector internal components out of tolerance	- Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest - Check and install new a fuel injector as required
P02D9-00	Cylinder 7 Fuel Injector Offset Learning at Max Limit - No sub type information	- Corrected set point voltage of the piezo actuator violates the on board diagnostic limit - Fuel injector internal components out of tolerance	- Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest - Check and install new a fuel injector as required
P02DB-00	Cylinder 8 Fuel Injector Offset Learning at Max Limit - No sub type information	- Corrected set point voltage of the piezo actuator violates the on board diagnostic limit - Fuel injector internal components out of tolerance	- Check for other related DTCs and refer to the relevant DTC index. Clear DTC and retest - Check and install new a fuel injector as required
P02EE-17	Cylinder 1 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE:	This DTC is set when the engine control module monitors a voltage on the cylinder 1 injector control circuit that is above the diagnostic threshold. The injector control circuit consists of a twisted pair of wires between the

		Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install new a fuel injector as required
P02EE-1C	Cylinder 1 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 1 fuel injector
P02EE-68	Cylinder 1 Injector Circuit Range /Performance - Event information	▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Corrupt engine control module software flash ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Clear the DTC and recheck the system ▪ If the DTC resets check and install new engine control module
P02EF-17	Cylinder 2 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module measured a	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short

		voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 2 fuel injector
P02EF-1C	Cylinder 2 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 2 fuel injector
P02EF-68	Cylinder 2 Injector Circuit Range /Performance - Event information	▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Corrupt engine control module software flash ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Clear the DTC and recheck the system ▪ If the DTC resets check and install new engine control module
P02F0-17	Cylinder 3 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 3 fuel injector

P02F0-1C	Cylinder 3 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23	■ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ■ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ■ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ■ Repair wiring harness as required. Check and install new cylinder 3 fuel injector
P02F0-68	Cylinder 3 Injector Circuit Range /Performance - Event information	■ Engine control module power supply failure ■ Engine control module ground supply failure ■ Corrupt engine control module software flash ■ Engine control module failure	■ Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit ■ Using the manufacturer approved diagnostic system re-configure the engine control module ■ Clear the DTC and recheck the system ■ If the DTC resets check and install new engine control module
P02F1-17	Cylinder 4 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12	■ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold ■ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ■ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ■ Repair wiring harness as required. Check and install new cylinder 4 fuel injector
P02F1-1C	Cylinder 4 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12	■ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ■ The injector control circuit consists of a twisted pair of wires between the

		▪ The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 4 fuel injector
P02F1-68	Cylinder 4 Injector Circuit Range /Performance - Event information	▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Corrupt engine control module software flash ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Clear the DTC and recheck the system ▪ If the DTC resets check and install new engine control module
P02F2-17	Cylinder 5 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 5 fuel injector
P02F2-1C	Cylinder 5 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module has detected a voltage outside of the	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to

		expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 5 fuel injector
P02F2-68	Cylinder 5 Injector Circuit Range /Performance - Event information	▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Corrupt engine control module software flash ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit ▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Clear the DTC and recheck the system ▪ If the DTC resets check and install new engine control module
P02F3-17	Cylinder 6 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 6 fuel injector
P02F3-1C	Cylinder 6 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range ▪ The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector ▪ Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits ▪ Repair wiring harness as required. Check and install new cylinder 6 fuel injector

P02F3-68	Cylinder 6 Injector Circuit Range /Performance - Event information	- Engine control module power supply failure - Engine control module ground supply failure - Corrupt engine control module software flash - Engine control module failure	- Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system re-configure the engine control module - Clear the DTC and recheck the system - If the DTC resets check and install new engine control module
P02F4-17	Cylinder 7 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold - The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector - Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits - Repair wiring harness as required. Check and install new cylinder 7 fuel injector
P02F4-1C	Cylinder 7 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 - The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range - The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector - Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits - Repair wiring harness as required. Check and install new cylinder 7 fuel injector
P02F4-68	Cylinder 7 Injector Circuit Range /Performance - Event information	- Engine control module power supply failure - Engine control module ground supply failure - Corrupt engine control module software flash	- Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system re-configure the engine control module

		Engine control module failure	- Clear the DTC and recheck the system - If the DTC resets check and install new engine control module
P02F5-17	Cylinder 8 Injector Circuit Range /Performance - Circuit voltage above threshold	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engine control module monitors a voltage on the injector control circuit that is above the diagnostic threshold - The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector - Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits - Repair wiring harness as required. Check and install new cylinder 8 fuel injector
P02F5-1C	Cylinder 8 Injector Circuit Range /Performance - Circuit voltage out of range	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 - The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engine control module monitors a voltage on the injector control circuit that is out of range - The injector control circuit consists of a twisted pair of wires between the engine control module and the Piezo actuator within the injector - Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits - Repair wiring harness as required. Check and install new cylinder 8 fuel injector
P02F5-68	Cylinder 8 Injector Circuit Range /Performance - Event information	- Engine control module power supply failure - Engine control module ground supply failure - Corrupt engine control module software flash - Engine control module failure	- Refer to the electrical circuit diagrams and check engine control module power and ground circuits for open circuit - Using the manufacturer approved diagnostic system re-configure the engine control module - Clear the DTC and recheck the system - If the DTC resets check and install new engine control module
P0300-00	Random Misfire Detected - No		Check for cylinder mis-fire, glow plug and injector DTCs and refer to the

	sub type information	▪ Fuel injector circuit failure(s) (injector DTCs also flagged) ▪ Fuel system failure	relevant DTC index. Refer to the electrical circuit diagrams and check injector circuits for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ▪ Check for fuel system failure. Repair harness as required. Clear DTC and retest
P0301-00	Cylinder 1 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ▪ Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0302-00	Cylinder 2 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ▪ Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0303-00	Cylinder 3 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ▪ Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0304-00	Cylinder 4 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ▪ Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index

			Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0305-00	Cylinder 5 Misfire Detected - No sub type information	- Fuel injector electrical circuit failure(s) (injector DTCs also flagged) - Fuel injector failure - Cylinder compression low	- Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest - Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index - Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0306-00	Cylinder 6 Misfire Detected - No sub type information	- Fuel injector electrical circuit failure(s) (injector DTCs also flagged) - Fuel injector failure - Cylinder compression low	- Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest - Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index - Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0307-00	Cylinder 7 Misfire Detected - No sub type information	- Fuel injector electrical circuit failure(s) (injector DTCs also flagged) - Fuel injector failure - Cylinder compression low	- Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest - Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index - Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0308-00	Cylinder 8 Misfire Detected - No sub type information	- Fuel injector electrical circuit failure(s) (injector DTCs also flagged) - Fuel injector failure - Cylinder compression low	- Refer to the electrical circuit diagrams and check injector circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest - Check for cylinder mis-fire, glow plug and injector DTCs and refer to the relevant DTC index - Check for fuel injector failure or blockage. Carry out cylinder compression tests
P0336-29	Crankshaft Position Sensor	 NOTE:	Refer to the electrical circuit diagrams and check crankshaft position sensor

	A Circuit Range /Performance - Signal invalid	Circuit reference CPS ▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, disconnected ▪ Crankshaft position sensor circuit shielding failure ▪ Crankshaft Position sensor failure ▪ Crankshaft position sensor foreign matter on sensor face, gap incorrect ▪ Target wheel failure	circuit for short circuit to ground, short circuit to power, open circuit, disconnected. Repair harness as required. Clear DTC and retest ▪ Check and install new crankshaft position sensor circuit shielding as required ▪ Check crankshaft position sensor for foreign matter on crankshaft position sensor face. Check crankshaft position sensor air gap ▪ Check and install new crankshaft position sensor as required. Check and install new target wheel as required
P0336-31	Crankshaft Position Sensor A Circuit Range /Performance - No signal	 NOTE: Circuit reference CPS ▪ Harness failure - Crankshaft position sensor circuits ▪ Crankshaft position sensor failure ▪ Crankshaft position sensor or reference target positioning incorrect	▪ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to the sensor, check the signal circuit for open circuits, short circuit to power, and short circuit to ground. Repair harness as required. Clear DTC and retest ▪ Refer to the relevant section of the workshop manual. Check the sensor and crankshaft target for damage, contamination, and correct mounting ▪ Check and install new crankshaft position sensor as required. Check and install new target wheel as required
P0336-64	Crankshaft Position Sensor A Circuit Range /Performance - Signal plausibility failure	 NOTE: Circuit reference CPS ▪ Crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, disconnected	▪ Refer to the electrical circuit diagrams and check crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, disconnected. Repair harness as required. Clear DTC and retest ▪ Check and install new crankshaft position sensor circuit shielding as required ▪ Check crankshaft position sensor for foreign matter on crankshaft position sensor face. Check crankshaft position sensor air gap

		■ Crankshaft position sensor circuit shielding failure ■ Crankshaft Position sensor failure ■ Crankshaft position sensor foreign matter on sensor face, gap incorrect ■ Target wheel failure	■ Check and install new crankshaft position sensor as required. Check and install new target wheel as required
P0341-31	Camshaft Position Sensor A Circuit Range /Performance (Bank 1 or single sensor) - No signal	 NOTE: Circuit reference CID ■ Harness failure - Camshaft position sensor circuits ■ Camshaft position sensor failure ■ Camshaft position sensor or reference target positioning incorrect	■ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to the sensor, check the signal circuit for open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Clear DTC and retest ■ Refer to the relevant section of the workshop manual. Check the sensor and camshaft target for damage, contamination, and correct mounting ■ Check and install new camshaft position sensor as required
P0341-3A	Camshaft Position Sensor A Circuit Range /Performance (Bank 1 or single sensor) - Incorrect has too many pulses	 NOTE: Circuit reference CID ■ Harness failure - Camshaft position sensor circuits ■ Camshaft position sensor failure ■ Camshaft position sensor or reference target positioning incorrect	■ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to the sensor, check the signal circuit for open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Clear DTC and retest ■ Refer to the relevant section of the workshop manual. Check the sensor and camshaft target for damage, contamination, and correct mounting. Check camshaft timing is to specification ■ Check and install new camshaft position sensor as required
P0342-64	Camshaft Position Sensor A Circuit Low (Bank 1 or single sensor) - Signal plausibility failure	■ Harness failure - Camshaft position sensor circuit ■ Camshaft position sensor failure ■ Camshaft position sensor or reference target positioning incorrect	■ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to the sensor, check the signal circuit for open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Clear DTC and retest

			- Check the sensor and camshaft target for damage, contamination, and correct mounting. Check camshaft timing is to specification - Check and install new camshaft position sensor as required
P0380-11	Glow Plug /Heater Circuit A - Circuit short to ground	 NOTE: Circuit reference GPC	- Using the manufacturer approved diagnostic system check datalogger signal, Glow Plug Coil Duty Cycle (0x9A04). Refer to the electrical circuit diagrams and check the control circuit from the engine control module to the glow plug control module for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new glow plug control module as required
P0380-12	Glow Plug /Heater Circuit A - Circuit short to battery	 NOTE: Circuit reference GPC	- Using the manufacturer approved diagnostic system check datalogger signal, Glow Plug Coil Duty Cycle (0x9A04). Refer to the electrical circuit diagrams and check the control circuit from the engine control module to the glow plug control module for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new glow plug control module as required

P0380-13	Glow Plug /Heater Circuit A - Circuit open	 NOTE: Circuit reference GPC	- Using the manufacturer approved diagnostic system check datalogger signal, Glow Plug Coil Duty Cycle (0x9A04). Refer to the electrical circuit diagrams and check the control circuit from the engine control module to the glow plug control module for short circuit to ground. Repair harness as required. Clear DTC and retest - Check and install new glow plug control module as required
P0380-4B	Glow Plug /Heater Circuit A - Over temperature	- The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure between engine control module and glow plug control module - Open circuit - Component failure - Glow plug control module failure	Using the manufacturer approved diagnostic system check datalogger signal, Glow Plug Coil Duty Cycle (0x9A04). Refer to the electrical circuit diagrams and check the glow plug heater circuit A for short circuit to ground, short circuit to power, high resistance. Repair harness as required. Clear DTC and retest
P0401-00	Exhaust Gas Recirculation A Flow Insufficient Detected - No sub type information	- Intake air system, low pressure boost leak - Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected - Exhaust gas recirculation actuator failure	- If this DTC is logged with P00BE-07 & P006A-00, suspect intake air system, low pressure boost leak - Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation actuator as required
P0402-00	Exhaust Gas Recirculation A Flow Excessive Detected - No sub type information	- Intake air system, high pressure boost leak - Exhaust gas recirculation actuator circuit short circuit to power, high resistance, open circuit, disconnected	- If this DTC is logged with P1247-00, P006A-00 & P00BF-07, suspect intake air system, high pressure boost leak - Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to power, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest

		Exhaust gas recirculation actuator failure	Check and install new exhaust gas recirculation actuator as required
P0403-13	Exhaust Gas Recirculation A Control Circuit - Circuit open	 NOTE: Circuit reference EGR_A_NEG EGR_A_POS - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Exhaust gas recirculation actuator circuit high resistance, open circuit, disconnected - Exhaust gas recirculation actuator failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB) - Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation control circuit for open circuit. This circuit consists of two wires connected between the engine control module and the exhaust gas recirculation valve motor. Check both wires for open circuit. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation control actuator as required
P0403-16	Exhaust Gas Recirculation A Control Circuit - Circuit voltage below threshold	 NOTE: Circuit reference EGR_A_NEG EGR_A_POS - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected - Exhaust gas recirculation actuator failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB) - Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation actuator as required
P0403-19	Exhaust Gas Recirculation A Control Circuit - Circuit	 NOTE:	Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB). Refer to the electrical circuit

	current above threshold	Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has measured current flow above a specified range ▪ Exhaust gas recirculation actuator circuit short circuit to power ▪ Exhaust gas recirculation actuator failure	diagrams and check exhaust gas recirculation actuator circuit for short circuit to power. Repair harness as required. Clear DTC and retest ▪ Check and install new exhaust gas recirculation actuator as required
P0403-1D	Exhaust Gas Recirculation A Control Circuit - Circuit current out of range	 NOTE: Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance, disconnected. Repair harness as required. Clear DTC and retest ▪ Check and install new exhaust gas recirculation actuator as required
P0403-4B	Exhaust Gas Recirculation A Control Circuit - Over temperature	 NOTE: Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module detected an internal temperature above the expected range ▪ Exhaust gas recirculation actuator	▪ Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance, disconnected. Repair harness as required. Clear DTC and retest ▪ Check and install new exhaust gas recirculation actuator as required

		circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected Exhaust gas recirculation actuator failure	
P0404-19	Exhaust Gas Recirculation A Control Circuit Range /Performance - Circuit current above threshold	 NOTE: Circuit reference EGR_A_NEG EGR_A_POS - The engine control module has measured current flow above a specified range - Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance - Exhaust gas recirculation actuator failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Bank 1 - Commanded (0x03FB) - Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation actuator as required
P0405-00	Exhaust Gas Recirculation Sensor A Circuit Low - No sub type information	 NOTE: Circuit reference EVP_A - Exhaust gas recirculation actuator sensor circuit short circuit to ground, high resistance, open circuit, disconnected - Exhaust gas recirculation actuator failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Valve Position Bank 1 (0x052E) - Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator sensor circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation actuator as required
P0405-11	Exhaust Gas Recirculation Sensor A Circuit Low - Circuit short to ground	 NOTE: Circuit reference EVP_A The engine control module has detected a	- Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation sensor signal circuit for short circuit to ground. Check the power and ground circuits which supply the sensor. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation sensor as required

		ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Exhaust gas recirculation actuator position sensor circuit short circuit to ground - Exhaust gas recirculation actuator failure	
P0406-00	Exhaust Gas Recirculation Sensor A Circuit High - No sub type information	 NOTE: Circuit reference EVP_A - Exhaust gas recirculation actuator circuit short circuit to power - Exhaust gas recirculation actuator failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Valve Position Bank 1 (0x052E) - Refer to the electrical circuit diagrams and check exhaust gas recirculation valve bank 1 circuits for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation sensor as required
P0406-12	Exhaust Gas Recirculation Sensor A Circuit High - Circuit short to battery	 NOTE: Circuit reference EVP_A - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Exhaust gas recirculation actuator position sensor circuit short circuit to power - Exhaust gas recirculation actuator failure	- Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation sensor signal circuit for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation sensor as required
P0409-16	Exhaust Gas Recirculation	 NOTE:	Refer to the electrical circuit diagrams and check exhaust gas recirculation

	Sensor A Circuit - Circuit voltage below threshold	Circuit reference EVP_A ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected ■ Exhaust gas recirculation actuator failure	actuator sensor circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest ■ Check and install new exhaust gas recirculation actuator as required
P0409-17	Exhaust Gas Recirculation Sensor A Circuit - Circuit voltage above threshold	 NOTE: Circuit reference EVP_A ■ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ■ Exhaust gas recirculation actuator circuit short circuit to power ■ Exhaust gas recirculation actuator failure	■ Refer to the electrical circuit diagrams and check exhaust gas recirculation actuator sensor circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new exhaust gas recirculation actuator as required
P040C-00	Exhaust Gas Recirculation Temperature Sensor A Circuit Low - No sub type information	 NOTE: Circuit reference EGRCIT ■ Pre-catalyst temperature sensor circuit, short circuit to ground ■ Pre-catalyst temperature sensor mechanical damage due to chaffing or heat	■ Refer to the electrical circuit diagrams and check pre-catalyst temperature sensor circuit for short circuit to ground ■ Check pre-catalyst temperature sensor for mechanical damage due to chaffing or heat ■ Check and install new pre-catalyst temperature sensor as required

P040D-00	Exhaust Gas Recirculation Temperature Sensor A Circuit High - No sub type information	 NOTE: Circuit reference EGRCIT - Pre-catalyst temperature sensor circuit, short circuit to power, open circuit - Pre-catalyst temperature sensor mechanical damage due to chaffing or heat	- Refer to the electrical circuit diagrams and check pre-catalyst temperature sensor circuit for short circuit to power, open circuit - Check pre-catalyst temperature sensor for mechanical damage due to chaffing or heat - Check and install new pre-catalyst temperature sensor as required
P041C-00	Exhaust Gas Recirculation Temperature Sensor B Circuit Low - No sub type information	 NOTE: Circuit reference EGRCOT Post-catalyst temperature sensor circuit, short circuit to ground	Refer to the electrical circuit diagrams and check post-catalyst temperature sensor circuit, for short circuit to ground
P041D-00	Exhaust Gas Recirculation Temperature Sensor B Circuit High - No sub type information	 NOTE: Circuit reference EGRCOT - Post-catalyst temperature sensor circuit, short circuit to power, open circuit - Post-catalyst temperature sensor mechanical damage due to chaffing or heat	- Refer to the electrical circuit diagrams and check post-catalyst temperature sensor circuit for short circuit to power, open circuit - Check post-catalyst temperature sensor for mechanical damage due to chaffing or heat - Check and install new post-catalyst temperature sensor as required
P0426-00	Catalyst Temperature Sensor Circuit Range /Performance (Bank 1, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference CCCIT_A Exhaust gas temperature sensor pre catalytic converter	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required

		circuit, short circuit to ground, open circuit, high resistance Exhaust gas temperature sensor pre catalytic converter failure	
P0427-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference CCCIT_A - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to ground - Exhaust gas temperature sensor pre catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required
P0428-00	Catalyst Temperature Sensor Circuit High (Bank 1, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference CCCIT_A - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor pre catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to power, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required
P042B-00	Catalyst Temperature Sensor Circuit Range /Performance (Bank1, Sensor Circuit 2) - No sub type information	 NOTE: Circuit reference CCCOT_A Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground, open circuit, high resistance - Check and install new exhaust gas temperature sensor post catalytic converter as required

		Exhaust gas temperature sensor post catalytic converter failure	
P042C-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 2) - No sub type information	 NOTE: Circuit reference CCCOT_A - Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground, open circuit, high resistance - Check and install new exhaust gas temperature sensor post catalytic converter as required
P042D-00	Catalyst Temperature Sensor Circuit High (Bank 1, Sensor Circuit 2) - No sub type information	 NOTE: Circuit reference CCCOT_A - Exhaust gas temperature sensor post catalytic converter short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to power, open circuit, high resistance - Check and install new exhaust gas temperature sensor post catalytic converter as required
P0435-00	Catalyst Temperature Sensor Circuit (Bank 2, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power - Exhaust gas temperature sensor	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to power - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to ground - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for open circuit - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for high resistance

		post turbocharger circuit, short circuit to ground - Exhaust gas temperature sensor post turbocharger circuit, open circuit - Exhaust gas temperature sensor post turbocharger circuit, high resistance - Exhaust gas temperature sensor post turbocharger circuit, disconnected - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for disconnected - Check and install new exhaust gas temperature sensor post turbocharger as required
P0436-00	Catalyst Temperature Sensor Circuit Range /Performance (Bank 2, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power, short circuit to ground, open circuit - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to power, short circuit to ground, open circuit - Check and install new exhaust gas temperature sensor post turbocharger as required
P0436-16	Catalyst Temperature Sensor Circuit Range /Performance (Bank 2, Sensor Circuit 1) - Circuit voltage below threshold	 NOTE: Circuit reference STOT - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas temperature sensor post turbocharger circuit, short circuit ground	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post turbocharger as required

		Exhaust gas temperature sensor post turbocharger failure	
P0437-00	Catalyst Temperature Sensor Circuit Low (Bank 2, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger circuit, short circuit to ground - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post turbocharger required
P0438-00	Catalyst Temperature Sensor Circuit High (Bank 2, Sensor Circuit 1) - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to power - Check and install new exhaust gas temperature sensor post turbocharger as required
P0480-11	Fan 1 Control Circuit - Circuit short to ground	 NOTE: Circuit reference ECFC The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected	- Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for short circuit to ground. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Clear DTC and retest - Check and install new cooling fan 1 control module as required

		▪ Fan 1 control circuit - Short circuit to ground ▪ Cooling fan 1 component failure	
P0480-12	Fan 1 Control Circuit - Circuit short to battery	 NOTE: Circuit reference ECFC ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Fan 1 control circuit - Short circuit to power ▪ Cooling fan 1 component failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for short circuit to power. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 1 control module as required
P0480-13	Fan 1 Control Circuit - Circuit open	 NOTE: Circuit reference ECFC ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Fan 1 control circuit - Open circuit ▪ Cooling fan 1 component failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for open circuit. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 1 control module as required
P0480-16	Fan 1 Control Circuit - Circuit voltage below threshold	 NOTE: Circuit reference ECFC	▪ Refer to the electrical circuit diagrams and check the fan 1 supply circuit for short circuit to ground, open circuit, high resistance. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 1 control module as required

		▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ The engine control module has determined that the fan 1 supply voltage is below the calibrated threshold ▪ Fan 1 control circuit - Open circuit, short circuit to ground, high resistance ▪ Cooling fan 1 control module failure	
P0480-17	Fan 1 Control Circuit - Circuit voltage above threshold	 NOTE: Circuit reference ECFC ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ The engine control module has determined that the fan 1 supply voltage is above the calibrated threshold ▪ Fan 1 control circuit - Short circuit to power ▪ Cooling fan 1 control module failure	▪ Refer to the electrical circuit diagrams and check the fan 1 supply circuit for short circuit to power. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 1 control module as required
P0480-71	Fan 1 Control Circuit - Actuator stuck	▪ Engine cooling fan partially stalled ▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by obstruction in fan cowling	▪ Confirm if customer has been deep water wading ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Using the manufacturer approved diagnostic system check datalogger signal - Electric Fan PWM Control - Commanded - (0x03F9) ▪ Using the manufacturer approved diagnostic system, operate cooling fan 1 through the full operating range and check that the DTC does not reset

P0480-97	Fan 1 Control Circuit - Component or system operation obstructed or blocked	- Engine cooling fan stalled - Engine cooling fan stalling caused by deep water wading - Engine cooling fan stalling caused by obstruction in fan cowling	- Confirm if customer has been deep water wading - Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required - Using the manufacturer approved diagnostic system check datalogger signal - Electric Fan PWM Control - Commanded - (0x03F9) - Using the manufacturer approved diagnostic system, operate cooling fan 1 through the full operating range and check that the DTC does not reset
P0481-11	Fan 2 Control Circuit - Circuit short to ground	 NOTE: Circuit reference ECFC_2 - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Fan 2 control circuit - Short circuit to ground - Cooling fan 2 component failure	- Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrical circuit diagrams and check the cooling fan 2 control circuit for short circuit to ground. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Clear DTC and retest - Check and install new cooling fan 2 control module as required
P0481-12	Fan 2 Control Circuit - Circuit short to battery	 NOTE: Circuit reference ECFC_2 - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Fan 2 control circuit - Short circuit to power - Cooling fan 2 component failure	- Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrical circuit diagrams and check the cooling fan 2 control circuit for short circuit to power. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Clear DTC and retest - Check and install new cooling fan 2 control module as required

P0481-13	Fan 2 Control Circuit - Circuit open	 NOTE: Circuit reference ECFC_2 ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Fan 2 control circuit - Open circuit ▪ Cooling fan 2 component failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrical circuit diagrams and check the cooling fan 2 control circuit for open circuit. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 2 control module as required
P0481-16	Fan 2 Control Circuit - Circuit voltage below threshold	 NOTE: Circuit reference ECFC_2 ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ The engine control module has determined that the fan 2 supply voltage is below the calibrated threshold ▪ Fan 2 control circuit - Open circuit, short circuit to ground, high resistance ▪ Cooling fan 2 control module failure	▪ Refer to the electrical circuit diagrams and check the fan 2 supply circuit for short circuit to ground, open circuit, high resistance. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 2 control module as required
P0481-17	Fan 2 Control Circuit - Circuit voltage above threshold	 NOTE: Circuit reference ECFC_2 ▪ The engine control module measured a	▪ Refer to the electrical circuit diagrams and check the fan 2 supply circuit for short circuit to power. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Clear DTC and retest ▪ Check and install new cooling fan 2 control module as required

		voltage above a specified range but not necessarily a short circuit to power ▪ The engine control module has determined that the fan 2 supply voltage is above the calibrated threshold ▪ Fan 2 control circuit - Short circuit to power ▪ Cooling fan 2 control module failure	
P0481-71	Fan 2 Control Circuit - Actuator stuck	▪ Engine cooling fan partially stalled ▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by obstruction in fan cowl	▪ Confirm if customer has been deep water wading ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Using the manufacturer approved diagnostic system check datalogger signal - Viscous Fan PWM Control - Commanded - (0x03FA) ▪ Using the manufacturer approved diagnostic system, operate cooling fan 2 through the full operating range and check that the DTC does not reset
P0481-97	Fan 2 Control Circuit - Component or system operation obstructed or blocked	▪ Engine cooling fan stalled ▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by obstruction in fan cowl	▪ Confirm if customer has been deep water wading ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Using the manufacturer approved diagnostic system check datalogger signal - Viscous Fan PWM Control - Commanded - (0x03FA) ▪ Using the manufacturer approved diagnostic system, operate cooling fan 2 through the full operating range and check that the DTC does not reset
P0486-16	Exhaust Gas Recirculation Sensor B Circuit - Circuit voltage below threshold	 NOTE: Circuit reference TPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground	▪ Refer to the electrical circuit diagrams and check exhaust gas recirculation valve B circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness as required. Clear DTC and retest ▪ Check and install new exhaust gas recirculation valve B as required

		- Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, open circuit, disconnected - Exhaust gas recirculation valve bank 2 failure	
P0486-17	Exhaust Gas Recirculation Sensor B Circuit - Circuit voltage above threshold	 NOTE: Circuit reference TPS - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas recirculation valve bank 2 circuit short circuit to power - Exhaust gas recirculation valve bank 2 failure	- Refer to the electrical circuit diagrams and check exhaust gas recirculation valve bank 2 circuit for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation valve bank 2 as required
P0487-00	Exhaust Gas Recirculation Throttle Control Circuit A / Open - No sub type information	 NOTE: Circuit reference TPA_POS TPA_NEG - Harness failure - Throttle motor circuit open circuit, short circuit to ground, short circuit to power - Intake air shut off throttle failure	- Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). This DTC is set when the engine control module detects an open load error on the throttle motor circuit. Refer to the electrical circuit diagrams and check the throttle control circuits for open circuit, short circuit to ground, short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new throttle motor actuator as required
P0487-19	Exhaust Gas Recirculation Throttle Control Circuit A / Open - Circuit current above threshold	 NOTE: Circuit reference TPA_POS TPA_NEG	Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). Refer to the electrical circuit diagrams and check the throttle control circuits for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest

		▪ The engine control module has measured current flow above a specified range ▪ Harness failure - Throttle motor circuit short circuit to ground, high resistance, open circuit ▪ Intake air shut off throttle failure	▪ Check and install new throttle motor actuator as required
P0488-16	Exhaust Gas Recirculation Throttle Control Circuit A Range /Performance - Circuit voltage below threshold	 NOTE: Circuit reference TPA_POS TPA_NEG ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Throttle motor circuit open circuit, short circuit to ground, short circuit to power, high resistance, short circuit to another circuit ▪ Intake air shut off throttle failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). Refer to the electrical circuit diagrams and check the throttle plate actuator control circuits for open circuit, short circuit to ground, short circuit to power, high resistance, short circuit to other circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new throttle motor actuator as required
P0488-19	Exhaust Gas Recirculation Throttle Control Circuit A Range /Performance - Circuit current above threshold	 NOTE: Circuit reference TPA_POS TPA_NEG ▪ The engine control module has measured current flow above a specified range ▪ Harness failure - Throttle motor circuit short circuit to ground, high resistance ▪ Intake air shut off throttle failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). Refer to the electrical circuit diagrams and check the throttle plate actuator control circuit for short circuit to ground, high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new throttle motor actuator as required
P0488-	Exhaust Gas	 NOTE:	

1D	Recirculation Throttle Control Circuit A Range /Performance - Circuit current out of range	Circuit reference TPA_POS TPA_NEG ■ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ■ Harness failure - Throttle motor circuit high resistance, open circuit, short circuit to ground, short circuit to power, short circuit to other circuit ■ Intake air shut off throttle failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). Refer to the electrical circuit diagrams and check the throttle plate actuator control circuits for high resistance, open circuit, short circuit to ground, short circuit to power, short circuit to other circuit. Repair harness as required. Clear DTC and retest ■ Check and install new throttle motor actuator as required
P0488-4B	Exhaust Gas Recirculation Throttle Control Circuit A Range /Performance - Over temperature	■ The engine control module detected an internal temperature above the expected range ■ Harness failure - Throttle motor circuit short circuit to ground, short circuit to power, high resistance ■ Intake air shut off throttle failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Commanded Throttle Actuator Control (0xF44C). Refer to the electrical circuit diagrams and check the throttle motor circuit for short circuit to ground, short circuit to power, high resistance. Repair harness as required. Clear DTC and retest ■ Check and install new throttle motor actuator as required
P0489-77	Exhaust Gas Recirculation Control Circuit Low - Commanded position not reachable	■ Harness failure - Exhaust gas recirculation electric actuator, short circuit to power ■ Harness failure - Exhaust gas recirculation electric actuator, short circuit to ground ■ Harness failure - Exhaust gas recirculation electric actuator, open circuit ■ Harness failure - Exhaust gas recirculation electric actuator, high resistance ■ Exhaust gas recirculation electric actuator failure of return spring	■ Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to power ■ Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to ground ■ Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for open circuit ■ Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for high resistance ■ Check exhaust gas recirculation electric actuator return spring ■ Check and install new exhaust gas recirculation electric actuator unit as required

		Exhaust gas recirculation electric actuator failure	
P0490-77	Exhaust Gas Recirculation A Control Circuit High - Commanded position not reachable	- Harness failure - Throttle EGR Inlet, short circuit to power - Harness failure - Throttle EGR Inlet, short circuit to ground - Harness failure - Throttle EGR Inlet, open circuit - Harness failure - Throttle EGR Inlet, high resistance - Throttle EGR Inlet failure	- Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to power - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to ground - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for open circuit - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for high resistance - Check and install new exhaust gas recirculation electric actuator unit as required
P049D-00	Exhaust Gas Recirculation A Control Position Exceeded Learning Limit - No sub type information	- Harness failure - Exhaust gas recirculation electric actuator, short circuit to power - Harness failure - Exhaust gas recirculation electric actuator, short circuit to ground - Harness failure - Exhaust gas recirculation electric actuator, open circuit - Harness failure - Exhaust gas recirculation electric actuator, high resistance - Exhaust gas recirculation electric actuator failure	- Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to power - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for short circuit to ground - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for open circuit - Refer to the electrical circuit diagrams and check exhaust gas recirculation electric actuator control circuits, for high resistance - Check and install new exhaust gas recirculation electric actuator unit as required
P0504-27	Brake Switch A / B Correlation - Signal rate of change above threshold	 NOTE: Circuit reference BOO_1 BOO_2 Brake pedal switch plunger failure	- Check for related brake pressure DTCs within the anti-lock braking control module - Check brake pedal switch is installed correctly. Clear DTC, start the engine and press the brake pedal, using maximum travel, for greater than 1 second taking care not to press the accelerator pedal

		▪ Brake pedal switch not installed correctly ▪ Brake switch A failure	▪ Check and install new brake switch as required
P0504-62	Brake Switch A / B Correlation - Signal compare failure	 NOTE: Circuit reference BOO_1 BOO_2 ▪ Error check for brake plausibility ▪ Brake pedal switch circuit short circuit to power, open circuit	▪ Refer to the electrical circuit diagrams and check brake pedal switch circuit for short circuit to power, open circuit. Repair harness as required. Clear DTC and retest
P050E-00	Cold Start Engine Exhaust Temperature Too Low - No sub type information	▪ This DTC is set when more than one exhaust gas temperature sensor is out of range or has a circuit failure. The engine control module diagnosis is unable to determine which sensor. The DTC may be set with other related exhaust gas temperature sensor DTCs	▪ Using the manufacturer approved diagnostic system, check for other exhaust gas temperature sensor related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check all other exhaust gas temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Repair wiring harness as required, clear DTC and retest ▪ Check and install new exhaust gas temperature sensor as required
P0512-64	Starter Request Circuit - Signal plausibility failure	 NOTE: Circuit reference CRNK ▪ Harness failure - Starter request circuit failure ▪ Central junction box failure ▪ Post drive cycle not completed ▪ Harness failure - CAN circuit	▪ This DTC is set when the engine control module detects the engine crank signal from the passenger compartment fuse box is stuck high. If this DTC is logged on its own clear the DTC using the manufacturer approved diagnostic system. Select PARK position. Set ignition OFF, wait 2 minutes for post drive to complete. Set ignition ON but do NOT crank. Wait 25 seconds. Crank engine and check if DTC is cleared. If this DTC is logged with lost communication DTCs refer to electrical circuit diagrams and check CAN circuit. If the engine cannot be cranked read the DTCs stored in the central junction box and refer to the relevant DTC index. Check the engine crank signal input circuit for open circuits, short circuit to power, short circuit to ground, short

			to other circuits. Repair wiring harness as required, clear DTC and retest system
P0513-00	Incorrect Immobilizer Key - No sub type information	Security key invalid	Check for CAN network interference /engine control module related error. Using the manufacturer approved diagnostic system, re-configure the engine control module. Using the manufacturer approved diagnostic system, re-configure the instrument panel control module
P0544-00	Exhaust Gas Temperature Sensor Circuit - Bank 1 Sensor 1 - No sub type information	 NOTE: Circuit reference CCCIT_A - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor pre catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required
P0547-00	Exhaust Gas Temperature Sensor Circuit - Bank 2 Sensor 1 - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger short circuit to ground - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required
P0548-00	Exhaust Gas Temperature Sensor Circuit Low - Bank 2 Sensor 1 - No sub type information	 NOTE: Circuit reference STOT Exhaust gas temperature sensor	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required

		post turbocharger short circuit to ground Exhaust gas temperature sensor post turbocharger failure	
P0549-00	Exhaust Gas Temperature Sensor Circuit High - Bank 2 Sensor 1 - No sub type information	 NOTE: Circuit reference STOT - Exhaust gas temperature sensor post turbocharger short circuit short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to power, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required
P0571-62	Brake Switch A Circuit - Signal compare failure	- The brake switch signal received over CAN is defective - Brake switch (footbrake switch) failure	- Using the manufacturer approved diagnostic system carry out network integrity test - Check for related DTCs and refer to the relevant DTC index - Check and install new brake switch as required
P0575-81	Cruise Control Input Circuit - Invalid serial data received	The DTC sets if the cancel, set minus, set plus, resume, headway increase and headway decrease speed control buttons have been pressed for longer than a calibrated period of time. The system then assumes a stuck /damaged button and will cancel and/or disable cruise. The failure will be healed in the next driving cycle if the failure is removed	Check speed control buttons are not jammed/contaminated/damaged. Check speed control module for related DTCs and refer to the relevant DTC index
P0602-00	Powertrain Control Module Programming Error - No sub type information	Mismatch between car configuration file and engine control module calibration for expected engine power output	Using the manufacturer approved diagnostic system re-configure the engine control module. Clear DTC and retest

		Incorrect engine control module calibration for vehicle specification	
P0605-00	Internal Control Module Read Only Memory (ROM) Error - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P0606-00	Control Module Processor - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P0606-44	Control Module Processor - Data memory failure	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P0606-49	Control Module Processor - Internal electronic failure	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P0607-00	Control Module Performance - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module

		Engine control module failure	power supply for open circuit. Repair harness as required. Clear DTC and retest Check and install new engine control module as required
P060A-00	Internal Control Module Monitoring Processor Performance - No sub type information	Engine control module has reset due to hardware failure	- Check engine control module power and ground supply circuits for open circuits. Re-configure the engine control module using the manufacturer approved diagnostic system, Clear DTC and retest - Check and install new engine control module as required
P060A-16	Internal Control Module Monitoring Processor Performance - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P060A-17	Internal Control Module Monitoring Processor Performance - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P060B-00	Internal Control Module A/D Processing Performance - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required

P060B-16	Internal Control Module A/D Processing Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P060B-17	Internal Control Module A/D Processing Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P060B-46	Internal Control Module A/D Processing Performance - Calibration /parameter memory failure	▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P060B-49	Internal Control Module A/D Processing Performance - Internal electronic failure	▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest

		Engine control module failure	- Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P060B-63	Internal Control Module A/D Processing Performance - Circuit /component protection time-out	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P060C-00	Internal Control Module Main Processor Performance - No sub type information	Engine control module has reset due to hardware or software failure	Clear DTC and road test vehicle including at least 5 ignition cycles. Re-check DTC and repeat above procedure. If DTC resets check and install new engine control module as required
P060D-00	Internal Control Module Accelerator Pedal Position Performance - No sub type information	- Accelerator pedal position sensor circuit, short circuit to ground, short circuit to power, high resistance, open circuit - Accelerator pedal position sensor failure - Engine control module failure	- Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Clear DTC and retest - Check and install an accelerator pedal position sensor as required - Check and install new engine control module as required
P0610-00	Control Module Vehicle Options Error - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest

			Check and install new engine control module as required
P0615-00	Starter Relay Circuit - No sub type information	 NOTE: Circuit reference SM_COMMAND_H SM_COMMAND_L Starter relay circuit failure	Refer to the electrical circuit diagrams and check starter relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance
P061A-00	Internal Control Module Torque Performance - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supplies for open circuit. Repair wiring harness as required, clear DTC and retest system - Refer to the electrical circuit diagrams and check the engine control module ground supply's for open circuit. Repair wiring harness as required, clear DTC and retest system - Check and install new engine control module as required
P061B-00	Internal Control Module Torque Calculation Performance - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply's for open circuit. Repair wiring harness as required, clear DTC and retest system - Refer to the electrical circuit diagrams and check the engine control module ground supply's for open circuit. Repair wiring harness as required, clear DTC and retest system - Check and install new engine control module as required
P061C-00	Internal Control Module Engine RPM Performance - No sub type information	- Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest

			- Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P0627-13	Fuel Pump A Control Circuit / Open - Circuit open	 NOTE: Circuit reference LPPR - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Wiring harness failure between engine control module and central junction box - Lift pump relay failure	 NOTE: When this DTC is set the engine control module limits engine torque to protect the engine and avoid pulling air into the fuel system under high load - Check the operation of the lift pump relay. Replace if required. Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junction box for open circuit. Repair harness as required. Clear DTC and retest - Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump
P0628-11	Fuel Pump A Control Circuit Low - Circuit short to ground	 NOTE: Circuit reference LPPR - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Wiring harness failure between engine control module and central junction box - Lift pump relay failure	 NOTE: When this DTC is set the fuel lift pump will run continuously when the ignition is on, even when the engine is stopped - Check the operation of the lift pump relay. Replace if required. Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junction box for a short circuit to ground. Repair harness as required. Clear DTC and retest - Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump
P0629-12	Fuel Pump A Control Circuit High - Circuit short to battery	 NOTE:	 NOTE:

		Circuit reference LPPR ■ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ■ Wiring harness failure between engine control module and central junction box ■ Lift pump relay failure	When this DTC is set the engine control module limits engine torque to protect the engine and avoid pulling air into the fuel system under high load ■ Check the operation of the lift pump relay. Replace if required. Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junction box for a short circuit to power. Repair harness as required. Clear DTC and retest ■ Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump
P062A-92	Fuel Pump A Control Circuit Range /Performance - Performance or incorrect operation	■ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ■ Lift pump relay has been driven by the central junction box, but has not been requested by the engine control module ■ Lift pump relay has been requested by the engine control module, but has not been driven by the central junction box ■ Wiring harness failure lift pump monitor circuit ■ Lift pump relay	■ Check for related DTCs within the central junction box ■ Refer to the electrical circuit diagrams and check fuel lift pump relay circuit for short circuit to ground, short circuit to power, open circuit. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check between the engine control module and the rear junction box for short circuit to ground. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check the rear junction box to in tank fuel pump relay for short circuit to power. Clear DTC and retest ■ Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump ■ Check and install new fuel lift pump relay as required. Clear DTC and retest
P062B-41	Internal Control Module Fuel Injector Control Performance - General checksum failure	■ Injector failure(s) ■ Corrupt engine control module software flash ■ Engine control module power supply failure ■ Engine control module ground supply failure ■ Engine control module failure	■ Check for related injector DTCs and repair these first. Clear DTCs and retest ■ Using the manufacturer approved diagnostic system re-configure the engine control module ■ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair

			harness as required. Clear DTC and retest Check and install new engine control module as required
P062B-48	Internal Control Module Fuel Injector Control Performance - Supervision software failure	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-49	Internal Control Module Fuel Injector Control Performance - Internal electronic failure	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-62	Internal Control Module Fuel Injector Control Performance - Signal compare failure	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest

			- Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-64	Internal Control Module Fuel Injector Control Performance - Signal plausibility failure	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-67	Internal Control Module Fuel Injector Control Performance - Signal incorrect after event	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-91	Internal Control Module Fuel Injector Control Performance - Parametric	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module

			power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-92	Internal Control Module Fuel Injector Control Performance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-94	Internal Control Module Fuel Injector Control Performance - Unexpected operation	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module - Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P062B-9A	Internal Control Module Fuel Injector Control Performance - Component or	- Injector failure(s) - Corrupt engine control module software flash - Engine control module power supply failure	- Check for related injector DTCs and repair these first. Clear DTCs and retest - Using the manufacturer approved diagnostic system re-configure the engine control module

	system operating conditions	▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P062D-00	Fuel Injector Driver Circuit Performance Bank 1 - No sub type information	▪ Bank 1 injector circuit (s) high resistance, short circuit to ground, short circuit to power, open circuit, disconnected ▪ Injector bank 1 failure	▪ Check for related injector DTCs. Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Clear DTC and retest ▪ Check and install new injector(s) as required
P062D-11	Fuel Injector Driver Circuit Performance Bank 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Wiring harness failure - Short circuit to ground	▪ Refer to the electrical circuit diagrams and check the wiring harness between engine control module and bank 1 fuel injectors for short circuit to ground
P062D-16	Fuel Injector Driver Circuit Performance Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Bank 1 injector circuit (s) high resistance, short circuit to ground, high resistance, open circuit, disconnected ▪ Injector bank 1 failure	▪ Check for related injector DTCs. Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits for short circuit to ground, open circuit, high resistance. Clear DTC and retest ▪ Check and install new injector(s) as required
P062D-17	Fuel Injector Driver Circuit Performance Bank 1 - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power	▪ Check for related injector DTCs. Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits for short circuit to power. Clear DTC and retest ▪ Check and install new injector(s) as required

		- Bank 1 injector circuit (s) short circuit to power - Injector bank 1 failure	
P062E-00	Fuel Injector Driver Circuit Performance Bank 2 - No sub type information	- Bank 2 injector circuit (s) high resistance, short circuit to ground, short circuit to power, high resistance, open circuit, disconnected - Injector bank 2 failure	- Check for related injector DTCs. Check injector connections to bank 2 cylinders. Refer to the electrical circuit diagrams and check bank 2 injector circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Clear DTC and retest - Check and install new injector(s) as required
P062E-11	Fuel Injector Driver Circuit Performance Bank 2 - Circuit short to ground	- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Wiring harness failure - Short circuit to ground	Refer to the electrical circuit diagrams and check the wiring harness between engine control module and bank 2 fuel injectors for short circuit to ground
P062E-17	Fuel Injector Driver Circuit Performance Bank 2 - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Bank 2 injector circuit (s) short circuit to power - Injector bank 2 failure	- Check for related injector DTCs. Check injector connections to bank 2 cylinders. Refer to the electrical circuit diagrams and check bank 2 injector circuits for short circuit to power. Clear DTC and retest - Check and install new injector(s) as required
P0633-00	Immobilizer Key Not Programmed - ECM/PCM - No sub type information	Security key not programmed	Check for CAN network interference /engine control module related errors. Re-configure the engine control module using the manufacturer approved diagnostic system. Re-configure the instrument panel control module using the manufacturer approved diagnostic system
P0634-1B	PCM / ECM / TCM Internal Temperature A Too High - Circuit resistance above threshold	- Engine control module internal temperature too high - Engine control module failure	- Clear DTC, wait 10 minutes and retest - Check and install new engine control module as required
P064D-	Internal		

00	Control Module O2 Sensor Processor Performance - Bank 1 - No sub type information	▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure ▪ Engine control module failure	▪ Check and repair related heated oxygen sensor DTCs. Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, short circuit to power, high resistance, open circuit ▪ Check and install new heated oxygen sensor as required. Clear DTC and retest ▪ Check and install new engine control module as required
P064D-16	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Heated oxygen sensor circuit short circuit to ground, high resistance, open circuit ▪ Heated oxygen sensor failure ▪ Engine control module failure	▪ Check and repair related heated oxygen sensor DTCs. Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit ▪ Check and install new heated oxygen sensor as required. Clear DTC and retest ▪ Check and install new engine control module as required
P0658-00	Actuator Supply Voltage A Circuit Low - No sub type information	▪ Actuator supply circuit voltage below threshold	▪ Using the manufacturer approved diagnostic system perform the (Turbo, EGR and air path dynamic test) routine ▪ Refer to the electrical circuit diagrams and check engine control module power and ground supplies for short circuit to ground, open circuit
P0659-00	Actuator Supply Voltage A Circuit High - No sub type information	▪ Actuator supply circuit voltage above threshold	▪ Using the manufacturer approved diagnostic system perform the (Turbo, EGR and air path dynamic test) routine ▪ Refer to the electrical circuit diagrams and check engine control module power and ground supplies for short circuit to power
P0667-16	PCM / ECM / TCM Internal Temperature Sensor A Range /Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine control module internal temperature too low ▪ Engine control module failure	▪ Clear DTC, wait 10 minutes and re-check ▪ Check and install new engine control module as required

P0668-00	PCM / ECM / TCM Internal Temperature Sensor A Circuit Low - No sub type information	Engine control module internal temperature sensor failure	Check and install new engine control module as required
P0669-00	Control Module Internal Temperature Sensor "A" Circuit High - No sub type information	Engine control module internal temperature sensor failure	Check and install new engine control module as required
P066B-12	Cylinder 1 Glow Plug Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug control circuit short circuit to power - Glow plug failure	- Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system - Check and install new glow plug as required
P066D-12	Cylinder 2 Glow Plug Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug control circuit short circuit to power - Glow plug failure	- Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system - Check and install new glow plug as required
P066F-12	Cylinder 3 Glow Plug Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug control circuit short circuit to power - Glow plug failure	- Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system - Check and install new glow plug as required

P0671-00	Cylinder 1 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0672-00	Cylinder 2 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0673-00	Cylinder 3 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0674-00	Cylinder 4 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0675-00	Cylinder 5 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required

P0676-00	Cylinder 6 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0677-00	Cylinder 7 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P0678-00	Cylinder 8 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check glow plug connector is correctly connected, clear DTC and retest system ▪ Check and install new glow plug as required
P067B-12	Cylinder 4 Glow Plug Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug control circuit short circuit to power ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system ▪ Check and install new glow plug as required
P067D-12	Cylinder 5 Glow Plug Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has	▪ Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system

		detected a vehicle power measurement when another value was expected ▪ Glow plug control circuit short circuit to power ▪ Glow plug failure	▪ Check and install new glow plug as required
P067F-12	Cylinder 6 Glow Plug Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Glow plug control circuit short circuit to power ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system ▪ Check and install new glow plug as required
P0683-00	Glow Plug Control Module to PCM Communication Circuit - No sub type information	 NOTE: Circuit reference GPC ▪ Engine control module to glow plug control module circuit failure ▪ Glow plug control module failure ▪ Engine control module failure	▪ This DTC is set when the data transmission on the control circuit between the engine control module and the glow plug control module has an error. Check for other related DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the glowplug control signal line from the engine control module to the glow plug control module for failures such as intermittent open circuits, high resistance, short circuit to ground, short circuit to power. Check the glowplug monitor signal line from the glowplug control module to the engine control module for intermittent open circuits, high resistance, short circuit to ground, short circuit to power ▪ Refer to the electrical circuit diagrams and check power and ground supplies to the glowplug control module. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P068A-00	ECM/PCM Power Relay De-Energized - Too Early - No sub type information	 NOTE: Circuit reference PWRSTN	▪ This DTC is set when the engine management system high current relay contacts open early - Indicating a power hold failure. Check the vehicle battery has not been disconnected before the engine management system relay has powered down. Check the operation of the engine

		- Vehicle battery disconnected before engine management system relay has powered down - Engine management system high current relay failure - Engine junction box failure - Harness failure - Relay control circuit	management system high current relay. Refer to the electrical circuit diagrams and check the engine management system high current relay supply and control circuits for open circuits, high resistance, short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Clear DTC and retest Check and install new relay or engine junction box as required
P068D-12	Cylinder 7 Glow Plug Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug control circuit short circuit to power - Glow plug failure	- Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system - Check and install new glow plug as required
P068F-12	Cylinder 8 Glow Plug Control Circuit High - Circuit short to battery	- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Glow plug control circuit short circuit to power - Glow plug failure	- Refer to the electrical circuit diagrams and check glow plug circuit for short circuit to power. Repair wiring harness as required, clear DTC and retest system - Check and install new glow plug as required
P06AF-00	Torque Management System - Forced Engine Shutdown - No sub type information	- Injection cut off demand for shut off coordinator - The DTC sets when the engine speed has exceeded the hard engine speed limiter. When this DTC sets, a request to the shut-off coordinator is made for a reversible shut-off of the injection power stages, ensuring no injections take place. It is reset when engine	Check for related engine control module DTCs and repair these first. Clear the DTC and retest

		speed falls less than or equal to the hard limiter ▪ Engine run away ▪ Excessive injected fuel ▪ Over boost from a turbocharger	
P0700-00	Transmission Control System (MIL Request) - No sub type information	▪ MIL request by automatic gearbox ▪ HS CAN network, short circuit to power, short circuit to ground, open circuit	▪ Check transmission control module for DTCs and refer to the relevant DTC index. Clear DTC and retest ▪ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit
P0850-64	Park / Neutral Switch Input Circuit - Signal plausibility failure	 NOTE: Circuit reference PNS ▪ Hardwired park/neutral switch does not match the information within the CAN signal ▪ Park/neutral switch circuit, short circuit to power, short circuit to ground, open circuit ▪ HS CAN network failure	▪ Refer to the electrical circuit diagrams and check the park/neutral switch circuit, for short circuit to power, short circuit to ground, open circuit ▪ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit
P0A09-16	DC/DC Converter Status Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P0A0F-00	Engine Failed to Start - No sub type information	▪ Engine control module has logged an extended crank with no engine start	▪ Check for related engine control module DTCs and repair these first. Clear DTC and retest

		▪ Harness failure - Starter motor circuit and relays ▪ Vehicle battery failure ▪ Starter motor mechanical failure	▪ Refer to the electrical circuit diagrams and check the starter motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check Starter motor for mechanical failure. Install a new starter motor as required
POA0F-71	Engine Failed to Start - Actuator stuck	▪ Vehicle battery failure ▪ Jammed starter motor	▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check for a jammed or damaged starter motor. Install a new starter motor as required
POA0F-97	Engine Failed to Start - Component Or System Operation Obstructed Or Blocked	 NOTE: To diagnose when starter motor activation results in no engine rotation and / or safety monitoring has terminated activation ▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Engine control module has logged an extended crank with no engine start ▪ Other related DTCs ▪ Harness failure - Starter motor circuit and relays circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Vehicle battery failure ▪ Starter motor jammed ▪ Starter motor mechanical failure	▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the starter motor circuit and relays circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check starter motor for mechanical integrity ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P0A10-17	DC/DC Converter Status Circuit High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check the engine control module power supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P0A14-13	Engine Mount Control A Circuit / Open - Circuit open	 NOTE: Circuit reference AEM_1 ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Wiring harness failure - Active engine mount solenoid circuit	▪ Refer to the electrical circuit diagrams and check the engine mount A control circuit for open circuit. Repair harness as required. Clear DTC and retest
P0A15-11	Engine Mount Control A Circuit Low - Circuit short to ground	 NOTE: Circuit reference AEM_1 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Wiring harness failure - Active engine mount solenoid circuit	▪ Refer to the electrical circuit diagrams and check the engine mount A control circuit for short circuit to ground. Repair harness as required. Clear DTC and retest

P0A16-12	Engine Mount Control A Circuit High - Circuit short to battery	 NOTE: Circuit reference AEM_1	Refer to the electrical circuit diagrams and check the engine mount A control circuit for short circuit to power. Repair harness as required. Clear DTC and retest
		- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Wiring harness failure - Active engine mount solenoid circuit	
P0A16-4B	Engine Mount Control A Circuit High - Over temperature	- The engine control module detected an internal temperature above the expected range - Active engine mount solenoid circuit short circuit to ground, short circuit to power - Active engine mount solenoid failure	- Refer to the electrical circuit diagrams and check active engine mount solenoid circuit for short circuit to ground, power. Repair harness as required. Clear DTC and retest - Check and install new active engine mount solenoid as required
P0A1A-00	Generator Control Module - No sub type information	- Generator to gateway control module LIN circuit, open circuit - The engine control module has not received the expected CAN signal from the gateway control module within the specified time interval - CAN harness link between engine control module and gateway control module network malfunction	- Check for good/clean contact at generator and gateway control module LIN circuit connectors/pins. Refer to the electrical circuit diagrams and check generator to gateway control module LIN circuit for open circuit. Check for gateway control module hardware DTCs and refer to relevant DTC index. Clear DTCs and repeat automated diagnostic procedure using the manufacturer approved diagnostic system - Using the manufacturer approved diagnostic system, check gateway control module for DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check gateway control module power and ground circuits for open circuit - Check CAN harness between engine control module and gateway control module, repair as necessary

P0A94-00	DC/DC Converter Performance - No sub type information	- Engine control module 5 volt power supply to sensors short circuit to power, short circuit to ground, open circuit - Engine control module failure	- Refer to the electrical circuit diagrams and check the engine control module 5 volt power supply to sensors for short circuit to power, short circuit to ground, open circuit. Repair wiring harness as required, clear DTC and retest system - Check and install new engine control module as required
P115A-00	Low Fuel Level - Forced Limited Power - No sub type information	- Low level fuel condition - Fuel level sensor signal circuit - Short circuit to ground, short circuit to power, open circuit - Fuel level sensor failure	- Check the fuel level, add fuel if required. Clear DTC and retest - If there is sufficient fuel, refer to the electrical circuit diagrams and check the fuel level sensor circuits for short, open circuit. Repair harness as required. Clear DTC and retest - Check and install new fuel level sensor as required
P115B-00	Low Fuel Level - Forced Engine Shutdown - No sub type information	- Low level fuel condition enabling run dry strategy - Fuel level sensor signal circuit - Short circuit to ground, short circuit to power, open circuit - Fuel level sensor failure	- Check the fuel level, add fuel if required. Clear DTC and retest - If there is sufficient fuel, refer to the electrical circuit diagrams and check the fuel level sensor circuits for short, open circuit. Repair harness as required. Clear DTC and retest - Check and install new fuel level sensor as required
P117B-00	Exhaust Gas Temperature Sensor Correlation - Bank 1 - No sub type information	This DTC is set when more than one exhaust gas temperature sensor is out of range or has a circuit failure. The engine control module diagnosis is unable to determine which sensor. The DTC may be set with other related exhaust gas temperature sensor DTCs	- Using the manufacturer approved diagnostic system, check for other exhaust gas temperature sensor related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check all other exhaust gas temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Repair wiring harness as required, clear DTC and retest - Check and install new exhaust gas temperature sensor as required
P120F-00	Fuel Pressure Regulator Excessive Variation - No sub type information	Fuel pressure regulator signal circuit, short circuit to power, short circuit to ground, open circuit	- Refer to the electrical circuit diagrams and check fuel pressure regulator circuit for short circuit to power, short circuit to ground, open circuit - Refer to the electrical circuit diagrams and check fuel pressure regulator

		▪ Fuel pressure regulator VREF circuit high resistance ▪ Fuel pressure regulator failure	VREF circuit for high resistance. Repair wiring harness as required, clear DTC and retest system ▪ Check for fuel pump related DTCs. Check and install new Fuel pressure regulator as required
P1247-00	Turbocharger Boost Pressure Low - No sub type information	▪ Air intake system, low pressure intake blocked or restricted ▪ Exhaust system blocked or restricted ▪ Boost air recirculation solenoid stuck open in bi-turbo mode ▪ Boost air solenoid stuck closed in bi-turbo mode ▪ Turbine intake solenoid stuck open ▪ Air intake system, boost air system high pressure boost air leak on bank 1 ▪ Air intake system, boost air system high pressure boost air leak in bi-turbo mode on bank 2 ▪ Variable geometry turbocharger compressor wheel seized	▪ Test air intake system for air leaks and check exhaust system for blockages or restriction ▪ If this DTC is logged with P00BE-00, suspect boost air recirculation solenoid stuck open in bi-turbo mode ▪ If this DTC is logged with P00BC-00, suspect boost air solenoid stuck closed in bi-turbo mode ▪ If this DTC is logged with P0235-94, P00BD-07, P22D2-77 & P22CF-71 suspect turbine intake solenoid stuck open ▪ If this DTC is logged with P006A-00, P0402-00 & P00BF-07 suspect air intake system, boost air system high pressure boost air leak on bank 1 ▪ If this DTC is logged with P006A-00 suspect air intake system, boost air system high pressure boost air leak in bi-turbo mode on bank 2 ▪ If this DTC is logged with P00BE-07 & P00BD-07 suspect variable geometry turbocharger compressor wheel seized ▪ If this DTC is logged with P0235-94, P00BD-07, P22D2-77 & P22CF-71 suspect Air intake system, low pressure intake blocked or restricted
P1395-13	Glow Plug Monitor Fault (Bank 1) - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Power feed from central junction box to glow control unit open circuit	▪ Refer to the electrical circuit diagrams and check Power feed from central junction box to glow control unit for open circuit
P1395-96	Glow Plug Monitor Fault (Bank 1) - Component internal failure	▪ Glow control unit, glow monitor circuit, short circuit to power, short circuit to ground, open circuit ▪ Glow control unit failure	▪ Refer to the electrical circuit diagrams and check glow control unit, glow monitor circuit, for short circuit to power, short circuit to ground, open circuit ▪ Check and install new glow control unit as required

P1396-13	Glow Plug Monitor Fault (Bank 2) - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Power feed from central junction box to glow control unit open circuit	▪ Refer to the electrical circuit diagrams and check Power feed from central junction box to glow control unit for open circuit
P1396-96	Glow Plug Monitor Fault (Bank 2) - Component internal failure	▪ Glow control unit, glow monitor 2 circuit, short circuit to power, short circuit to ground, open circuit ▪ Glow control unit failure	▪ Refer to the electrical circuit diagrams and check glow control unit, glow monitor 2 circuit, for short circuit to power, short circuit to ground, open circuit ▪ Check and install new glow control unit as required
P1551-32	Cylinder 1 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Injector control circuit failure ▪ Injector failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the injector control circuits for intermittent failures, open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Repair harness as required. Clear DTC and retest ▪ Refer to the workshop manual and check the injector is to specification and has been installed correctly. Check and install new injector as required ▪ Check and install new engine control module as required
P1551-35	Cylinder 1 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL11 O_P_PVH11 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value	▪ Refer to the electrical circuit diagrams and check the injector control circuits for intermittent failures, open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Repair harness as required. Clear DTC and retest ▪ Refer to the workshop manual and check the injector is to specification and has been installed correctly. Check and install new injector as required ▪ Check and install new engine control module as required

		▪ Injector control circuit failure ▪ Injector failure ▪ Engine control module failure	
P1552-32	Cylinder 2 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1552-35	Cylinder 2 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1553-32	Cylinder 3 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other

		▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	
P1553-35	Cylinder 3 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL23 O_P_PVH23 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1554-32	Cylinder 4 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1554-	Cylinder 4		

35	Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL12 O_P_PVH12	- Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power - Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1555-32	Cylinder 5 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21	- Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground - Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1555-35	Cylinder 5 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL21 O_P_PVH21	- Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power - Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other

		▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	
P1556-32	Cylinder 6 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1556-35	Cylinder 6 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1557-32	Cylinder 7 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other

		▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	
P1557-35	Cylinder 7 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL14 O_P_PVH14 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1558-32	Cylinder 8 Injector Circuit Range /Performance - Signal low time < minimum	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to ground ▪ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other

P1558-35	Cylinder 8 Injector Circuit Range /Performance - Signal high time > maximum	 NOTE: Circuit reference O_P_PVL24 O_P_PVH24 ■ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ■ Harness failure - fuel injector control circuit short circuit to power ■ Harness failure - fuel injector control circuit short to each other	■ Refer to the electrical circuit diagrams and check fuel injector control circuit for short circuit to power ■ Refer to the electrical circuit diagrams and check fuel injector control circuit for short to each other
P1575-62	Pedal Position Out Of Self Test Range - Signal compare failure	 NOTE: Circuit reference BOO_2 ■ The engine control module detected failure when comparing two or more input parameters for plausibility ■ Brake pedal switch circuit short circuit to power, open circuit ■ Brake pedal switch internal failure	■ Refer to the electrical circuit diagrams and check brake pedal switch circuit for short circuit to power, open circuit ■ Check and install new brake pedal switch as required ■ Clear the DTCs, drive the vehicle at greater than 11mph (17kph) with a throttle pedal greater than 10% for greater than 3 seconds. Press the brake pedal, using maximum travel for greater than 3 seconds, with the vehicle stationary, press the brake pedal using maximum travel for greater than 1 second
P1627-16	Module Supply Voltage Out Of Range - Circuit voltage below threshold	 NOTE: Circuit reference VBR1 VBR2 ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Vehicle battery failure ■ Charging system failure	■ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ■ Check the vehicle charging system performance to ensure the voltage regulation is correct ■ Using the manufacturer approved diagnostic system check datalogger signals, Control module voltage (0xF442). This DTC is set when the engine control module supply voltage is below the threshold value. Refer to the electrical circuit diagrams and check the four engine control module power ground circuits for high resistance or open circuits

		- Engine control module power or ground supply - Engine junction box failure - Power distribution failure - Engine control module failure	- Check the switched power supply feeds to the engine control module which come from the engine junction box through the engine management system relay. Repair harness as required. Clear DTC and retest - Check and install new engine control module as required
P1627-17	Module Supply Voltage Out Of Range - Circuit voltage above threshold	 NOTE: Circuit reference VBR1 VBR2 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Vehicle battery failure - Charging system failure - Engine control module power or ground supply - Central junction box failure - Power distribution failure	- Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests - Check the vehicle charging system performance to ensure the voltage regulation is correct - Using the manufacturer approved diagnostic system check datalogger signals, Control module voltage (0xF442). This DTC is set when the engine control module supply voltage is greater than the threshold value. Refer to the electrical circuit diagrams and check the four engine control module power ground circuits for high resistance or open circuits - Check the switched power supply feeds to the engine control module which come from the central junction box through the engine management system relay. Repair harness as required. Clear DTC and retest
P1631-73	Main Relay (power hold) - Actuator stuck closed	 NOTE: Circuit reference PWRSTN - Engine management system high current relay failure - Relay control circuit failure	- This DTC is set when the engine management system high current relay contacts are detected stuck closed by the engine control module. Check the operation of the engine management system high current relay. Refer to the electrical circuit diagrams and check the engine management system high current relay supply and control circuits for open circuits, high resistance, short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Clear DTC and retest - Check and install new relay as required
P165D-01	Grill Shutter "A" Control Circuit - General electrical failure	- Grill shutter actuator supply voltage falls outside the specified range - Grill shutter actuator internal circuit failure	Refer to the electrical circuit diagrams and check the grill shutter actuator circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Repair harness as required

			- Using the manufacturer approved diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset - Check and install new grill shutter actuator as required
P165D-07	Grill Shutter "A" Control Circuit - Mechanical failures	- Grill shutter "A" actuator has become disconnected from the grill shutter - Linkage between the grill shutter "A" actuator and grill shutter has failed	- Check all vanes within the grill shutter are connected to the grill shutter actuator. Repair as required. Clear DTC and retest - Using the manufacturer approved diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset - Check and install new grill shutter as required
P165D-13	Grill Shutter "A" Control Circuit - Circuit open	- Grill shutter "A" control circuit - Open circuit - Grill shutter actuator internal circuit failure	- Refer to the electrical circuit diagrams and check the grill shutter actuator control circuit for open circuit. Repair harness as required - Check and install new grill shutter actuator as required
P165D-4B	Grill Shutter "A" Control Circuit - Over temperature	The grill shutter actuator internal circuits are overheated	- Check the grill shutter for damage, obstructions and blockages. Repair as required. Clear DTC and retest - Using the manufacturer approved diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset - Check and install new grill shutter actuator as required
P165D-87	Grill Shutter "A" Control Circuit - Missing message	- The engine control module has not received the expected signal from the grill shutter "A" actuator - Grill shutter "A" control circuit - Open circuit - Grill shutter actuator internal circuit failure	- Refer to the electrical circuit diagrams and check the grill shutter actuator control circuit for open circuit, high resistance. Repair harness as required. Clear DTC and retest - Check and install new grill shutter actuator as required
P165D-97	Grill Shutter "A" Control Circuit - Component or system	- Grill shutter damaged, obstructed or blocked - Grill shutter actuator has failed all calibration attempts	- Check the grill shutter for damage, obstructions and blockages. Repair as required. Clear DTC and retest - Using the manufacturer approved diagnostic system, operate the grill shutter through the full operating

	operation obstructed or blocked		range and check that the DTC does not reset Check and install new grill shutter actuator as required
P1712-00	Transmission Torque Reduction Request Signal - No sub type information	Unintended torque request signal sent over CAN from transmission control module to engine control module. The engine control module recognises it is unintended and applies a torque limit and sets the DTC	Check transmission control module for related DTCs and refer to relevant DTC index, Clear DTC and retest
P1719-68	Engine Torque Signal - Event information	- Working limitation information - The DTC sets as a result of an engine torque limitation caused by an engine overheat situation	- Check engine control module for related DTCs and refer to relevant DTC index - Refer to the workshop manual and check the cooling system is functioning correctly
P2031-00	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 2 - No sub type information	 NOTE: Circuit reference CCCOT_A - Exhaust gas temperature sensor post catalytic converter short circuit to ground - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post catalytic converter as required
P2080-16	Exhaust Gas Temperature Sensor Circuit Range /Performance Bank 1 Sensor 1 - Circuit voltage below threshold	 NOTE: Circuit reference CCCIT_A The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground	This DTC is set when there is a plausibility error on the signal from the exhaust gas temperature sensor pre catalytic converter. Refer to the workshop manual and check the exhaust gas temperature sensor pre catalytic converter and wiring harness for obvious signs of mechanical damage due to chaffing or heat. The exhaust gas temperature sensor pre catalytic converter is a thermistor with a signal and ground connection. Refer to the electrical circuit diagrams and check the signal circuit for open circuit, short circuit to power, short circuit to ground, high resistance.

		■ Harness failure - Exhaust gas temperature sensor pre catalytic converter ■ Exhaust gas temperature sensor pre catalytic converter failure	Check the ground circuit for open circuit, high resistance, short circuit to power. Repair the wiring harness as required ■ Check and install new exhaust gas temperature sensor pre catalytic converter as required
P2082-00	Exhaust Gas Temperature Sensor Circuit Range /Performance Bank 2 Sensor 1 - No sub type information	 NOTE: Circuit reference STOT ■ Exhaust gas temperature sensor post turbocharger short circuit short circuit to ground, open circuit, high resistance ■ Exhaust gas temperature sensor post turbocharger failure	■ Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground, open circuit, high resistance ■ Clear the DTC and retest ■ Check and install new exhaust gas temperature sensor post turbocharger as required
P2084-16	Exhaust Gas Temperature Sensor Circuit Range /Performance Bank 1 Sensor 2 - Circuit voltage below threshold	 NOTE: Circuit reference CCCOT_A ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Exhaust gas temperature sensor post catalytic converter circuit, short circuit to ground ■ Exhaust gas temperature sensor post catalytic converter failure	■ Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground ■ Check and install new exhaust gas temperature sensor post catalytic converter as required
P2122-00	Throttle/Pedal Position Sensor /Switch D Circuit Low - No sub type information	 NOTE: Circuit reference APP_1	■ Using the manufacturer approved diagnostic system check datalogger signals, Accelerator Pedal Position D (0xF449), Pedal Position Sensor Voltage - Sensor 1 (0x0914). The Accelerator Pedal Position sensor consists of two potentiometer circuits feeding independent pedal demand

		■ Harness failure - Accelerator pedal position sensor circuit ■ Accelerator pedal position sensor failure	signals to the engine control module. This DTC is set when the engine control module detects the pedal demand 1 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new accelerator pedal position sensor as required
P2123-00	Throttle/Pedal Position Sensor /Switch D Circuit High - No sub type information	 NOTE: Circuit reference APP_1 ■ Harness failure - Accelerator pedal position sensor circuit ■ Accelerator pedal position sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Accelerator Pedal Position D (0xF449), Pedal Position Sensor Voltage - Sensor 1 (0x0914). The Accelerator Pedal Position sensor consists of two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control module detects the pedal demand 1 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new accelerator pedal position sensor as required
P2127-00	Throttle/Pedal Position Sensor /Switch E Circuit Low - No sub type information	 NOTE: Circuit reference APP_2 ■ Harness failure - Accelerator pedal position sensor circuit ■ Accelerator pedal position sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Accelerator Pedal Position E (0xF44A), Pedal Position Sensor Voltage - Sensor 2 (0x0915). The Accelerator Pedal Position sensor consists of two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control module detects the pedal demand 2 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new accelerator pedal position sensor as required

P2128-00	Throttle/Pedal Position Sensor /Switch E Circuit High - No sub type information	 NOTE: Circuit reference APP_2 ■ Harness failure - Accelerator pedal position sensor circuit ■ Accelerator pedal position sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Accelerator Pedal Position E (0xF44A), Pedal Position Sensor Voltage - Sensor 2 (0x0915). The Accelerator Pedal Position sensor consists of two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control module detects the pedal demand 2 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new accelerator pedal position sensor as required
P2138-00	Throttle/Pedal Position Sensor /Switch D / E Voltage Correlation - No sub type information	■ Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit ■ Accelerator pedal position sensor failure	■ Refer to the electrical circuit diagrams and check accelerator pedal position sensor 1 circuit for short circuit to ground, short circuit to power, open circuit. Refer to the electrical circuit diagrams and check accelerator pedal position sensor 2 circuit for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new accelerator pedal position sensor as required
P2141-00	Exhaust Gas Recirculation Throttle Control Circuit A Low - No sub type information	 NOTE: Circuit reference TPA_POS TPA_NEG ■ Harness failure - Throttle motor circuit ■ Throttle motor failure	■ This DTC is set when the engine control module detects a short circuit to ground on the throttle motor circuit. Refer to the electrical circuit diagrams and check both the throttle plate actuator control circuits for short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new throttle motor actuator as required
P2141-77	Exhaust Gas Recirculation Throttle Control Circuit A Low - Commanded position not reachable	 NOTE: Circuit reference TPA_POS TPA_NEG ■ Throttle motor circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check throttle motor circuit for short circuit to ground ■ Check and install new throttle motor actuator as required

		Throttle motor failure	
P2142-00	Exhaust Gas Recirculation Throttle Control Circuit A High - No sub type information	 NOTE: Circuit reference TPA_POS TPA_NEG - Harness failure - Throttle motor circuit - Throttle motor failure	- This DTC is set when the engine control module detects a short circuit to power on the throttle motor circuit. Refer to the electrical circuit diagrams and check both the Throttle Plate Actuator control circuits for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new throttle motor actuator as required
P2142-77	Exhaust Gas Recirculation Throttle Control Circuit A High - Commanded position not reachable	- Throttle motor circuit short circuit to power - Throttle motor failure	- Refer to the electrical circuit diagrams and check throttle motor circuit for short circuit to power - Check and install new throttle motor actuator as required
P2177-00	System Too Lean Off Idle - Bank 1 - No sub type information	- Oxygen concentration implausibly high - Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for excess fuel at exhaust manifold, downpipe, heated oxygen sensor - Check for fuel / injector related DTCs and repair these first. Clear DTC and retest - Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2178-00	System Too Rich Off Idle - Bank 1 - No sub type information	- Oxygen concentration implausibly low - Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor - Check for fuel / injector related DTCs and repair these first. Clear DTC and retest - Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2191-00	System Too Lean at Higher Load - Bank 1	Oxygen concentration implausibly high	Check for excess fuel at exhaust manifold, downpipe, heated oxygen sensor

	- No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for fuel / injector related DTCs and repair these first. Clear DTC and retest - Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2192-00	System Too Rich at Higher Load - Bank 1 - No sub type information	- Oxygen concentration implausibly low - Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor - Check for fuel / injector related DTCs and repair these first. Clear DTC and retest - Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2195-00	O2 Sensor Signal Biassed /Stuck Lean - Bank 1, Sensor 1 - No sub type information	- Air leak at exhaust manifold, downpipe, heated oxygen sensor bank 1 - Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for fuel / injector related DTCs and repair these first. Clear DTC and retest - Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor - Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2226-62	Barometric Pressure Sensor A Circuit - Signal compare failure	- Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Differential pressure sensor failure - The engine control module has been submersed in water or mud - The engine control module has been	- Refer to the electrical circuit diagrams and check differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check the engine control module is clean and dry - Check the engine control module is not sealed by any non approved covering - Clear DTC and retest - Check and install new differential pressure sensor as required

		sealed in a non approved covering ▪ Differential pressure sensor failure ▪ Engine control module failure	▪ Check and install new engine control module as required
P2228-00	Barometric Pressure Sensor A Circuit Low - No sub type information	▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check engine control module power supply circuits for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check engine control module ground supply circuits for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P2229-00	Barometric Pressure Sensor A Circuit High - No sub type information	▪ Corrupt engine control module software flash ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approved diagnostic system re-configure the engine control module ▪ Refer to the electrical circuit diagrams and check engine control module power supply circuits for open circuit. Repair harness as required. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check engine control module ground supply circuits for open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new engine control module as required
P2245-00	O2 Sensor Reference Voltage Circuit Low - Bank 1, Sensor 1 - No sub type information	 NOTE: Circuit reference LPV_A ▪ Harness failure - Heated oxygen sensor circuit ▪ Oxygen sensor failure	▪ This DTC is set if the engine control module detects the bank 1, heated oxygen sensor 1 reference voltage is lower than expected. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuits, high resistance, short circuit to ground. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P2246-00	O2 Sensor Reference Voltage Circuit	 NOTE:	▪ This DTC is set if the engine control module detects the bank 1, heated oxygen sensor 1 reference voltage is

	High - Bank 1, Sensor 1 - No sub type information	Circuit reference LPV_A ■ Harness failure - Heated oxygen sensor circuit ■ Oxygen sensor failure	greater than expected. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuits, high resistance, short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P2263-92	Turbocharger /Supercharger Boost System Performance - Performance or incorrect operation	■ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ■ Induction system air leak or blockage ■ Boost air system leak or blockage ■ Manifold Absolute Pressure sensor A failure ■ Variable geometry turbine vane actuator sticking, failure ■ Turbocharger A failure	■ Check induction system for leaks, blockages ■ Check boost air system for leaks, blockages. Check for related DTCs ■ Check and install new Manifold Absolute Pressure sensor as required ■ Check turbocharger rod connection and oil seals ■ Check and install new variable geometry turbine vane actuator as required. Check install new turbocharger as required
P2264-00	Water in Fuel Sensor Circuit - No sub type information	 NOTE: Circuit reference WIF ■ Fuel supply incorrect /contaminated ■ Harness failure - Water in fuel sensor circuit ■ Water in fuel sensor failure ■ Central junction box power distribution failure	■ This DTC is set if the engine control module receives information indicating an implausible fuel level. Refer to the workshop manual and check the fuel system to ensure there is an adequate fuel level in the tank and the system is not leaking fuel or suffering from air ingress. Check the fuel for contamination by other fluids (petrol or water etc) ■ Refer to the electrical wiring diagrams and check the water in fuel sensor signal circuit between the sensor and the engine control module for high resistance, open circuit, short circuit to ground, short circuit to power, intermittent failures. Repair harness as required ■ Check and install new water in fuel sensor ■ Refer to the electrical wiring diagrams and check the water in fuel sensor power supply from the central junction box for high resistance, open circuit, short circuit to ground. Check the water in fuel sensor ground supply circuit for high resistance, open circuit, short circuit to power. Check the voltage supply from the central junction box, check and replace fuses

			as required. Replace central junction box if there is no voltage supply to the water in fuel sensor
P2269-68	Water in Fuel Condition - Event information	- Water in fuel - Water in fuel sensor circuit short circuit to power, open circuit - Water in fuel sensor failure	- Drain water from fuel filter housing. Clear DTC and wait 30 seconds and re-check DTC has cleared - Refer to the electrical circuit diagrams and check water in fuel sensor circuit for short circuit to power, open circuit. Repair harness as required - Check and install new water in fuel sensor as required
P226B-00	Turbocharger /Supercharger Boost Pressure Too High - Mechanical - No sub type information	- Error path indicating whether over boost shut down is active or not - This DTC is set when overboost shut down is active. This means that the vehicle has lost control of boost. The engine is shut down as a safety precaution. The engine control module checks for an overboost by system monitoring and also monitors vehicle acceleration, if in gear and engine acceleration at idle, if in neutral	Check for turbocharging related DTCs and repair these first. Clear DTC and retest
P2297-00	O2 Sensor Out of Range During Deceleration Bank 1, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Heated oxygen sensor failure	- Check for fuel / injector related DTCs and repair these first. Clear DTC and retest. Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2299-68	Brake Pedal Position /Accelerator Pedal Position Incompatible - Event information	- Brake pedal switch - Circuit Brake_SW_1 - Short circuit to ground, open circuit - Brake pedal switch - Circuit Brake_SW_2 - Short circuit to ground - Brake pedal switch incorrect adjustment - Brake pedal switch failure	- Refer to the electrical circuit diagrams and check brake pedal switch - Circuit Brake_SW_1 - For short circuit to ground, open circuit. Repair harness as required - Refer to the electrical circuit diagrams and check brake pedal switch - Circuit Brake_SW_2 - For short circuit to ground - Repair wiring harness as required, clear DTC. Start the engine press the brake pedal, using maximum travel

		Brake pedal pressed by driver at same time as accelerator pedal pressed	for greater than 1 second taking care not to press the accelerator pedal. Check the system is operating correctly and the DTC does not return - Check and adjust brake switch as required - Check and install new brake pedal switch as required
P22C5-11	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit short to ground	 NOTE: Circuit reference CSOV - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Harness failure - Turbocharger boost air solenoid circuit short circuit to ground	Refer to the electrical circuit diagrams and check the turbocharger boost air solenoid circuit between the engine control module and the control valve for a short circuit to ground. Check power supply to the control valve. Repair harness as required. Clear DTC and retest
P22C5-12	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit short to battery	 NOTE: Circuit reference CSOV - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Harness failure - Turbocharger boost air solenoid circuit short circuit to power - Turbocharger boost air solenoid failure	- Refer to the electrical circuit diagrams and check the turbocharger boost air solenoid circuit between the engine control module and the control valve for a short circuit to power. Check power supply to the control valve. Repair harness as required. Clear DTC and retest - Check and install new turbocharger boost air solenoid as required
P22C5-13	Turbocharger Compressor Outlet Valve	 NOTE:	Refer to the electrical circuit diagrams and check the turbocharger boost air solenoid circuit between the engine

	Control Circuit / Open - Circuit open	Circuit reference CSOV ■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ■ Harness failure - Turbocharger boost air solenoid circuit open circuit ■ Turbocharger boost air solenoid failure	control module and the control valve for open circuit. Check power supply to the control valve. Repair harness as required. Clear DTC and retest ■ Check and install new turbocharger boost air solenoid as required
P22C5-4B	Turbocharger Compressor Outlet Valve Control Circuit / Open - Over temperature	■ The engine control module detected an internal temperature above the expected range ■ Harness failure - Turbocharger boost air solenoid circuit open circuit ■ Turbocharger boost air solenoid failure	■ Refer to the electrical circuit diagrams and check the turbocharger boost air solenoid circuit between the engine control module and the control valve for open circuit. Check power supply to the control valve. Repair harness as required. Clear DTC and retest ■ Check and install new turbocharger boost air solenoid as required. Check and install new turbocharger boost air solenoid as required
P22D4-16	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Circuit voltage below threshold	 NOTE: Circuit reference TSVP ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Turbine intake valve position sensor circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit short circuit to ground. Repair harness as required. Clear DTC and retest
P22D4-17	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Circuit voltage above threshold	 NOTE: Circuit reference TSVF	■ Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new turbine intake valve position sensor as required

		▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ This is the long term adaption limit diagnostic. It diagnoses that the adapted values for the actuator end stops is outside of tolerance. This could be caused by sensor drift over time ▪ Turbine intake valve position sensor circuit short circuit to power ▪ Turbine intake valve position sensor stuck	
P242A-00	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 3 - No sub type information	 NOTE: Circuit reference PFOT ▪ Harness failure - Exhaust gas temperature sensor post DPF ▪ Exhaust gas temperature sensor post DPF failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required ▪ Check and install new exhaust gas temperature sensor post DPF as required
P242B-16	Exhaust Gas Temperature Sensor Circuit Range /Performance Bank 1 Sensor 3 - Circuit voltage below threshold	 NOTE: Circuit reference PFOT ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground	▪ This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module ▪ Refer to the workshop manual and check the exhaust gas temperature sensor post DPF for obvious signs of damage ▪ Check the sensor harness for chaffing or heat damage ▪ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03F6), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03C8)

		■ Harness failure - Exhaust gas temperature sensor post DPF ■ Exhaust gas temperature sensor post DPF failure	■ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair wiring as required ■ Check and install new exhaust gas temperature sensor post DPF as required
P242B-64	Exhaust Gas Temperature Sensor Circuit Range /Performance Bank 1 Sensor 3 - Signal plausibility failure	 NOTE: Circuit reference PFOT ■ Harness failure - Exhaust gas temperature sensor post DPF ■ Exhaust gas temperature sensor post DPF failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required ■ Check and install new exhaust gas temperature sensor post DPF as required
P242C-00	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 3 - No sub type information	 NOTE: Circuit reference PFOT ■ Harness failure - Exhaust gas temperature sensor post DPF ■ Exhaust gas temperature sensor post DPF failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas temperature sensor post DPF signal voltage is less than the engine control module was expecting. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required ■ Check and install new exhaust gas temperature sensor post DPF as required
P242D-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 3 -	 NOTE: Circuit reference PFOT	■ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature

	Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas temperature sensor post DPF circuit, short circuit to power, open circuit ▪ Exhaust gas temperature sensor post DPF failure	Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6) ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF circuit, for short circuit to power, open circuit ▪ Check and install new exhaust gas temperature sensor post DPF as required
P244A-00	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - No sub type information	▪ Diagnostic failure check for minimum pressure differential characteristics ▪ Diesel particulate filter internal components are missing or destroyed	 NOTE: If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index ▪ Clear DTC and re-test ▪ Check and install new diesel particulate filter as required
P244B-68	Diesel Particulate Filter Differential Pressure Too High (Bank 1) - Event information	▪ Engine protection back pressure high ▪ Sudden increases in differential pressure across the diesel particulate filter	 NOTE: This DTC when logged on its own is advisory only and no further action should be taken ▪ Using the manufacturer approved diagnostic system, check for related DTCs. If this DTC exists with any other diesel particulate filter differential pressure sensor DTCs, follow the advice given for that DTC
P2452-95	Diesel Particulate Filter Pressure Sensor A Circuit - Incorrect assembly	 NOTE: In cold climates differential pressure sensor	 NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs

		hose lines or metal pipes may be frozen ■ Differential pressure sensor hoses connected incorrectly ■ Differential pressure sensor hoses crushed, blocked, split	carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa ■ Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). Refer to the workshop manual and check differential pressure sensor hoses are installed correctly ■ Check differential pressure sensor hoses for crushed, blockage, split
P2453-00	Diesel Particulate Filter Pressure Sensor A Circuit Range /Performance - No sub type information	 NOTE: Circuit reference DPS ■ Harness failure - Differential pressure sensor ■ Differential pressure sensor failure	 NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured'

ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor fails a plausibility check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required
- Check and install new differential pressure sensor as required

P2454-00

Diesel Particulate Filter Pressure Sensor A
Circuit Low - No sub type information



NOTE:

Circuit reference DPS

- Harness failure - Differential pressure sensor
- Differential pressure sensor crossed hose lines
- Differential pressure sensor failure



NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30

seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is less than the threshold set in the engine control module diagnostic check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required
- Check differential pressure sensor hose lines are installed correctly
- Check and install new differential pressure sensor as required

P2455-00

Diesel Particulate Filter Pressure Sensor A Circuit High - No sub type information



NOTE:

Circuit reference
DPS

- Harness failure - Differential pressure sensor
- Differential pressure sensor failure
- Exhaust back pressure is too high



NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2

minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approved diagnostic system, check engine control module, for related DTCs and refer to the relevant DTC index
- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is greater than the threshold set in the engine control module diagnostic check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to power, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required
- Check and install new differential pressure sensor as required

P2456-00

Diesel Particulate Filter Pressure Sensor A Circuit Intermittent /Erratic - No sub type information

- Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- In cold climate or off road driving differential pressure sensor hoses crushed, blocked, split
- Differential pressure sensor failure



NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured'

ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Refer to the electrical circuit diagrams and check differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- Check differential pressure sensor hoses for crushed, blocked, split
- Check and replace diesel particulate filter differential pressure sensor as required

P2458-66

Diesel Particulate Filter Regeneration Duration (Bank 1) - Signal has too many transitions /events

- Permanent regeneration



NOTE:

This code is enabled for Land Rover engineering detailed diagnostics only. No further action should be taken

- Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index

P2459-65

Diesel Particulate Filter Regeneration Frequency (Bank 1) - Signal has too few transitions /events

- Blocked regeneration
- Customer driving routine does not allow the system to clean the particulate filter



NOTE:

If DTC is P2459-65 or AMBER DPF FULL REFER TO HANDBOOK message is displayed with no other reported messages. No repair is required, if the vehicle is driven on a highway AS DIRECTED IN THE HANDBOOK then the light will be extinguished and the system self healed, nothing more than this is required

			- Refer to the diesel particulate filter regeneration procedure and carry out a diesel particulate filter regeneration - Advise customer of driving routine required to regenerate diesel particulate filter as stated in the vehicle handbook
P245A-11	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit short to ground	 NOTE: Circuit reference EGRCBV	Using the manufacturer approved diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrical circuit diagrams and check the exhaust gas recirculation bypass solenoid circuit for short circuit to ground. Check power supply to the solenoid. Repair harness as required. Clear DTC and retest
P245A-12	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit short to battery	 NOTE: Circuit reference EGRCBV	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrical circuit diagrams and check the exhaust gas recirculation bypass solenoid circuit for short circuit to power. Check power supply to the solenoid. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation bypass solenoid as required

P245A-13	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit open	 NOTE: Circuit reference EGRCBV - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure - Exhaust gas recirculation bypass solenoid circuit open circuit - Exhaust gas recirculation bypass solenoid failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrical circuit diagrams and check the exhaust gas recirculation bypass solenoid circuit for open circuit. Check power supply to the solenoid. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation bypass solenoid as required
P245B-19	Exhaust Gas Recirculation Cooler Bypass Control Circuit Range /Performance - Circuit current above threshold	 NOTE: Circuit reference EGRCBV - The engine control module has measured current flow above a specified range - Harness failure - Exhaust gas recirculation bypass solenoid circuit short circuit to ground, high resistance - Exhaust gas recirculation bypass solenoid failure	- Using the manufacturer approved diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrical circuit diagrams and check the exhaust gas recirculation bypass solenoid circuit for short circuit to ground, high resistance. Repair harness as required. Clear DTC and retest - Check and install new exhaust gas recirculation bypass solenoid as required
P2463-00	Diesel Particulate Filter Restriction - Soot Accumulation (Bank 1) - No sub type information	Maximum soot mass	Refer to the relevant pinpoint test in section 309-00
P250A-36	Engine Oil Level Sensor Circuit - Signal	 NOTE: 	Using the manufacturer approved diagnostic system check datalogger signals, Engine Oil Level - Measured

	frequency too low	Circuit reference OTL ▪ The engine control module detected excessive duration for one cycle of the output across a specified sample size ▪ Oil level and temperature sensor circuit short circuit to ground, high resistance, open circuit ▪ Oil level and temperature sensor failure	(0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperature - Measured (0x03F3). Refer to the electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new oil level and temperature sensor as required
P250A-37	Engine Oil Level Sensor Circuit - Signal frequency too high	 NOTE: Circuit reference OTL ▪ The engine control module detected insufficient duration for one cycle of the output across a specified sample size ▪ Oil level and temperature sensor circuit short circuit to power ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperature - Measured (0x03F3). Refer to the electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to power. Repair harness as required. Clear DTC and retest ▪ Check and install new oil level and temperature sensor as required
P250A-38	Engine Oil Level Sensor Circuit - Signal frequency incorrect	 NOTE: Circuit reference OTL ▪ Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperature - Measured (0x03F3). Refer to the electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new oil level and temperature sensor as required

P250A-47	Engine Oil Level Sensor Circuit - Watchdog /safety MicroController failure	 NOTE: Circuit reference OTL	- Using the manufacturer approved diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperature - Measured (0x03F3). Refer to the electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new oil level and temperature sensor as required
P250A-92	Engine Oil Level Sensor Circuit - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Oil level and temperature sensor failure	- Using the manufacturer approved diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperature - Measured (0x03F3). Refer to the electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest - Check and install new oil level and temperature sensor as required
P2562-32	Turbocharger Boost Control Position Sensor A Circuit - Signal low time < minimum	- This DTC is set when the engine control module detects that the signal low time is less than the minimum value - Variable geometry turbine vane actuator circuit short circuit to ground - Variable geometry turbine vane actuator failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground - Check and install new variable geometry turbine vane actuator as required
P2562-35	Turbocharger Boost Control Position Sensor A Circuit - Signal high time > maximum	This DTC is set when the engine control module detects that the signal high time is greater than the maximum value	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required

		- Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator failure	
P2562-36	Turbocharger Boost Control Position Sensor A Circuit - Signal frequency too low	- The engine control module detected excessive duration for one cycle of the output across a specified sample size - Variable geometry turbine vane actuator circuit short circuit to ground - Variable geometry turbine vane actuator failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to ground - Check and install new variable geometry turbine vane actuator as required
P2562-37	Turbocharger Boost Control Position Sensor A Circuit - Signal frequency too high	- The engine control module detected insufficient duration for one cycle of the output across a specified sample size - Variable geometry turbine vane actuator circuit short circuit to power - Variable geometry turbine vane actuator failure	- Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator circuit for short circuit to power - Check and install new variable geometry turbine vane actuator as required
P25A2-00	Brake System Control Module Requested MIL Illumination - No sub type information	- MIL request by anti-lock brake system control module - HS CAN network, short circuit to power, short circuit to ground, open circuit	- Check anti-lock brake system control module for DTCs and refer to the relevant DTC index. Clear DTC and retest - If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit
P268C-00	Cylinder 1 Injector Data Incompatible - No sub type information	- Injector calibration data held in the engine control module is different to that read from the injector - Injector calibration data not stored / programmed	Using the manufacturer approved diagnostic system reprogram the injector codes

P268C-51	Cylinder 1 Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268D-00	Cylinder 2 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268D-51	Cylinder 2 Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268E-00	Cylinder 3 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268E-51	Cylinder 3 Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268F-00	Cylinder 4 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P268F-	Cylinder 4		

51	Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2690-00	Cylinder 5 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2690-51	Cylinder 5 Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2691-00	Cylinder 6 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2691-51	Cylinder 6 Injector Data Incompatible - Not programmed	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2692-00	Cylinder 7 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approved diagnostic system reprogram the injector codes
P2692-	Cylinder 7		

51	Injector Data Incompatible - Not programmed	- Injector calibration data held in the engine control module is different to that read from the injector - Injector calibration data not stored / programmed	Using the manufacturer approved diagnostic system reprogram the injector codes
P2693-00	Cylinder 8 Injector Data Incompatible - No sub type information	- Injector calibration data held in the engine control module is different to that read from the injector - Injector calibration data not stored / programmed	Using the manufacturer approved diagnostic system reprogram the injector codes
P2693-51	Cylinder 8 Injector Data Incompatible - Not programmed	- Injector calibration data held in the engine control module is different to that read from the injector - Injector calibration data not stored / programmed	Using the manufacturer approved diagnostic system reprogram the injector codes
P2A00-16	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage below threshold	 NOTE: Circuit reference LPTR_A - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Harness failure - Heated oxygen sensor failure - Heated oxygen sensor failure	- This DTC is set when the engine control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is less than the voltage threshold. This may be caused by the heated oxygen sensor being too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all the heated oxygen sensor circuits for open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control module power and ground supplies. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P2A00-17	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage above threshold	 NOTE: Circuit reference LPTR_A The engine control module measured a	This DTC is set when the engine control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is greater than the voltage threshold. This may be caused by the heated oxygen sensor being too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical

		voltage above a specified range but not necessarily a short circuit to power ■ Harness failure - Heated oxygen sensor failure ■ Heated oxygen sensor failure	damage. Refer to the electrical circuit diagrams and check all the heated oxygen sensor circuits for open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control module power and ground supplies. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
U0001-88	High Speed CAN Communication Bus - Bus off	■ High speed CAN bus failure ■ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit	■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ■ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit
U0101-00	Lost Communication with TCM - No sub type information	■ The engine control module has not received the expected CAN signal from the transmission control module within the specified time interval ■ CAN harness link between engine control module and transmission control module network malfunction	■ Using the manufacturer approved diagnostic system, check transmission control module for DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transmission control module power and ground circuits for open circuit. Check CAN harness between engine control module and transmission control module, repair as necessary
U0102-00	Lost Communication with Transfer Case Control Module - No sub type information	■ The engine control module has not received the expected CAN signal from the transfer case control module within the specified time interval ■ CAN harness link between engine control module and transfer case control module network malfunction	■ Using the manufacturer approved diagnostic system, check transfer case control module for DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transfer case control module power and ground circuits for open circuit. Check CAN harness between engine control module and transfer case control module, repair as necessary
U0103-00	Lost Communication With Gear Shift Control Module A - No sub type information	■ The engine control module has not received the expected CAN signal from the transmission control switch within the specified time interval ■ CAN harness link between engine control module and transmission control	■ Using the manufacturer approved diagnostic system, check transmission control switch for DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transmission control switch power and ground circuits for open circuit. Check CAN harness between engine control

		switch network malfunction	module and transmission control switch, repair as necessary
U0104-00	Lost Communication With Cruise Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the speed control module within the specified time interval ▪ CAN harness link between engine control module and speed control module network malfunction	▪ Using the manufacturer approved diagnostic system, check speed control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check speed control module power and ground circuits for open circuit. Check CAN harness between engine control module and speed control module, repair as necessary
U0121-00	Lost Communication With Anti-Lock Brake System (ABS) Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	▪ Using the manufacturer approved diagnostic system, check anti-lock brake system control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti-lock brake system control module, repair as necessary
U0126-00	Lost Communication With Steering Angle Sensor Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the steering angle sensor control module within the specified time interval ▪ CAN harness link between engine control module and steering angle sensor control module network malfunction	▪ Using the manufacturer approved diagnostic system, check steering angle sensor control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check steering angle sensor control module power and ground circuits for open circuit. Check CAN harness between engine control module and steering angle sensor control module, repair as necessary
U0128-00	Lost Communication With Park Brake Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the parking brake control module within the specified time interval ▪ CAN harness link between engine control module and parking brake control module network malfunction	▪ Using the manufacturer approved diagnostic system, check parking brake control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check parking brake control module power and ground circuits for open circuit.

			Check CAN harness between engine control module and parking brake control module, repair as necessary
U012A-00	Lost Communication with Chassis Control Module "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the integrated suspension control module within the specified time interval ▪ CAN harness link between engine control module and integrated suspension control module network malfunction	▪ Using the manufacturer approved diagnostic system, check integrated suspension control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and integrated suspension control module, repair as necessary
U0131-00	Lost Communication With Power Steering Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the power steering control module within the specified time interval ▪ CAN harness link between engine control module and power steering control module network malfunction	▪ Using the manufacturer approved diagnostic system, check power steering control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check power steering control module power and ground circuits for open circuit. Check CAN harness between engine control module and power steering control module, repair as necessary
U0133-00	Lost Communication With Active Roll Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the dynamic response control module within the specified time interval ▪ CAN harness link between engine control module and dynamic response control module network malfunction	▪ Using the manufacturer approved diagnostic system, check dynamic response control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check dynamic response control module power and ground circuits for open circuit. Check CAN harness between engine control module and dynamic response control module, repair as necessary
U0138-00	Lost Communication with All Terrain Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the terrain response switchpack within the specified time interval ▪ CAN harness link between engine control	▪ Using the manufacturer approved diagnostic system, check terrain response switchpack for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check terrain response switchpack power

		module and terrain response switchpack network malfunction	and ground circuits for open circuit. Check CAN harness between engine control module and terrain response switchpack, repair as necessary
U0140-00	Lost Communication With Body Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the central junction box within the specified time interval ▪ CAN harness link between engine control module and central junction box network malfunction	▪ Using the manufacturer approved diagnostic system, check central junction box for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check central junction box power and ground circuits for open circuit. Check CAN harness between engine control module and central junction box, repair as necessary
U0146-00	Lost Communication With Gateway "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the gateway control module within the specified time interval ▪ CAN harness link between engine control module and gateway control module network malfunction	▪ Using the manufacturer approved diagnostic system, check gateway control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check gateway control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and gateway control module, repair as necessary
U0151-00	Lost Communication With Restraints Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ▪ CAN harness link between engine control module and restraints control module network malfunction	▪ Using the manufacturer approved diagnostic system, check restraints control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check restraints control module power and ground circuits for open circuit. Check CAN harness between engine control module and restraints control module, repair as necessary
U0151-08	Lost Communication With Restraints Control Module - Bus signal /message failures	▪ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ▪ CAN harness link between engine control module and restraints	▪ Using the manufacturer approved diagnostic system, check restraints control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check restraints control module power and ground circuits for open circuit. Check

		control module network malfunction	CAN harness between engine control module and restraints control module, repair as necessary
U0155-00	Lost Communication With Instrument Panel Cluster (IPC) Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the instrument cluster within the specified time interval ▪ CAN harness link between engine control module and instrument cluster network malfunction	▪ Using the manufacturer approved diagnostic system, check instrument cluster for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check instrument cluster power and ground circuits for open circuit. Check CAN harness between engine control module and instrument cluster, repair as necessary
U0159-00	Lost Communication With Parking Assist Control Module "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the park distance control module within the specified time interval ▪ CAN harness link between engine control module and park distance control module network malfunction	▪ Using the manufacturer approved diagnostic system, check park distance control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check park distance control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and park distance control module, repair as necessary
U0164-00	Lost Communication With HVAC Control Module - No sub type information	▪ CAN harness link between engine control module and automatic temperature control module network malfunction ▪ The engine control module has not received the expected CAN signal from the automatic temperature control module within the specified time interval	▪ Using the manufacturer approved diagnostic system, check automatic temperature control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check automatic temperature control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and automatic temperature control module, repair as necessary
U0167-00	Lost Communication With Vehicle Immobilizer Control Module - No sub type information	▪ Engine control module identity transfer failed	▪ Check the immobilizer control module for DTCs and refer to the relevant DTC index. Refer to the electrical circuit diagrams and check the power supply and ground connections to the immobilizer control module. Using the manufacturer approved diagnostic system, complete a CAN network

			integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required ▪ Check and install new immobilizer control module
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ CAN network failure ▪ Central junction box failure ▪ Car configuration file incorrect	▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Car configuration signal not received. Check central junction box for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system check and update the car configuration file as required
U0402-83	Invalid Data Received from TCM - Value of signal protection calculation incorrect	▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0404-68	Invalid Data Received from Gear Shift Control Module A - Event information	▪ CAN network failure ▪ Implausible CAN data received from transmission control switch	▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0405-68	Invalid Data Received From Cruise Control Module - Event information	▪ Speed control system failure	▪ Check speed control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check and install new speed control module

U0405-82	Invalid Data Received From Cruise Control Module - Alive /sequence counter incorrect / not updated	▪ Speed control system failure ▪ CAN network failure ▪ Harness failure - Speed control module power supply or ground failure	▪ Check speed control buttons are not jammed/contaminated/damaged. Check speed control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Check and install new speed control module
U0405-84	Invalid Data Received From Cruise Control Module - Sequence counter incorrect	▪ Speed control system failure	▪ Check speed control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check and install new speed control module
U0405-86	Invalid Data Received From Cruise Control Module - Signal invalid	▪ Speed control system failure	▪ Check speed control buttons are not jammed/contaminated/damaged. Check speed control module for related DTCs and refer to the relevant DTC index
U0407-00	Invalid Data Received From Glow Plug Control Module - No sub type information	▪ This DTC is set when the glow plug control module does not match what is expected by the engine control module. Detection of the coding word is complete when 2 of 3 coding words match. This process is a one time operation and will be completed during assembly of the vehicle	▪ Check the correct glow plug control module is installed to the vehicle
U0407-81	Invalid Data Received From	▪ Glow plug control system failure	▪ Refer to electrical circuit diagrams and check the power supply and ground

	Glow Plug Control Module - Invalid serial data received		connections to the glow plug control module. Check the diagnostic circuit between the engine control module and the glow plug control module for short circuit to power, short circuit to ground, open circuits. Clear the DTC and retest Check and install new glow plug control module
U0415-68	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Event information	Signal error for vehicle speed over CAN	Using the manufacturer approved diagnostic system, check anti-lock brake system control module for DTCs and refer to the relevant DTC index
U0416-46	Invalid Data Received From Vehicle Dynamics Control Module - Calibration /parameter memory failure	Anti-lock brake system failure	Using the manufacturer approved diagnostic system, check anti-lock brake system control module for DTCs and refer to the relevant DTC index
U0416-68	Invalid Data Received From Vehicle Dynamics Control Module - Event information	- The engine control module has received the default brake pressure signal value over CAN from the anti-lock brake system control module for a specified time interval - Anti-lock brake system failure - CAN harness link between engine control module and anti-lock brake system control module network malfunction	- Using the manufacturer approved diagnostic system, check anti-lock brake system control module for DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti-lock brake system control module, repair as necessary
U0416-92	Invalid Data Received From Vehicle Dynamics Control Module - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Difference between anti-lock brake system speed signal value and instrument cluster speed value at low vehicle speeds	Check the anti-lock brake system control module for DTCs and refer to the relevant DTC index
U0422-00	Invalid Data Received From	The engine control module has not	Using the manufacturer approved diagnostic system, check central

	Body Control Module - No sub type information	received the expected CAN signal from the central junction box within the specified time interval ■ CAN harness link between engine control module and central junction box network malfunction	junction box for DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ■ Refer to the electrical circuit diagrams and check central junction box power and ground circuits for open circuit ■ Check CAN harness between engine control module and central junction box, repair as necessary
U0424-00	Invalid Data Received From HVAC Control Module - No sub type information	■ The engine control module has not received the expected CAN signal from the automatic temperature control module within the specified time interval ■ CAN harness link between engine control module and automatic temperature control module network malfunction	■ Using the manufacturer approved diagnostic system, check automatic temperature control module for DTCs and refer to the relevant DTC index ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ■ Refer to the electrical circuit diagrams and check automatic temperature control module power and ground circuits for open circuit ■ Check CAN harness between engine control module and automatic temperature control module, repair as necessary
U0424-68	Invalid Data Received From HVAC Control Module - Event information	■ Automatic temperature control module failure ■ Power distribution system failure - fuse failure ■ CAN network failure	■ This DTC is set if the engine control module loses communication with the automatic temperature control module. Check the automatic temperature control module for DTCs and refer to the relevant DTC index ■ Refer to the electrical circuit diagrams and check the power supply and ground connections to the automatic temperature control module ■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required ■ If there are no failures in the wiring harness and network communications check and install new automatic temperature control module as required
U0426-00	Invalid Data Received From Vehicle Immobilizer Control Module - No sub type information	■ Immobilizer control module has received an invalid identity response ■ Module substituted	■ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required. Using the manufacturer approved diagnostic

			system check and update the car configuration file as required ▪ Ensure all modules installed in the vehicle which store vehicle identity are valid for this vehicle and are not substitutes from a donor vehicle
U1A14-00	CAN Initialization Failure - No sub type information	▪ Harness failure - CAN circuit failure	▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2005-64	Vehicle Speed - Signal plausibility failure	▪ This DTC is set when the engine control module has recognized a vehicle speed signal plausibility failure ▪ Anti-lock brake system failure	▪ Using the manufacturer approved diagnostic system, check anti-lock brake system control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check instrument cluster for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2005-84	Vehicle Speed - Signal below allowable range	▪ Anti-lock brake system failure	▪ This DTC is set when the engine control module has recognized a vehicle speed signal which is below the allowable range. Check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index. Check the instrument cluster for related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2005-85	Vehicle Speed - Signal above allowable range	▪ Anti-lock brake system failure	▪ This DTC is set when the engine control module has recognized a vehicle speed signal which is above the allowable range. Check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index. Check the instrument cluster for related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit

			diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-00	Adaptive Cruise Control - No sub type information	Adaptive speed control system failure - Error indicating adaptive speed control failure flag set	Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-24	Adaptive Cruise Control - Signal stuck high	- The engine control module measures a signal that remains high when transitions are expected - Adaptive speed control system failure - Adaptive speed control follow speed error	Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-64	Adaptive Cruise Control - Signal plausibility failure	Adaptive speed control system failure - Adaptive speed control follow speed range error	Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-68	Adaptive Cruise Control - Event information	Adaptive speed control system failure - Error indicating adaptive speed control follow speed check when stationary	Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-86	Adaptive Cruise Control - Signal invalid	Adaptive speed control system failure - Error when invalid adaptive speed control resume requests are present	Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required